

An aerial photograph of a city street grid. The buildings are modern, multi-story structures. Many of the buildings have green roofs, which are covered in various types of vegetation, including small trees and shrubs. The streets are lined with trees, and there are green spaces between the buildings. The overall scene is a mix of urban development and greenery.

SPATIAL JUSTICE THROUGH URBAN GREEN SPACE CONNECTIVITY

Applied Spatial Analysis for Sustainable Urban Development

Group E

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GUANGZHOU

population of 18.680.000

7.433,3 KM2

45,5 % green coverage

high density

historical core



DELFT

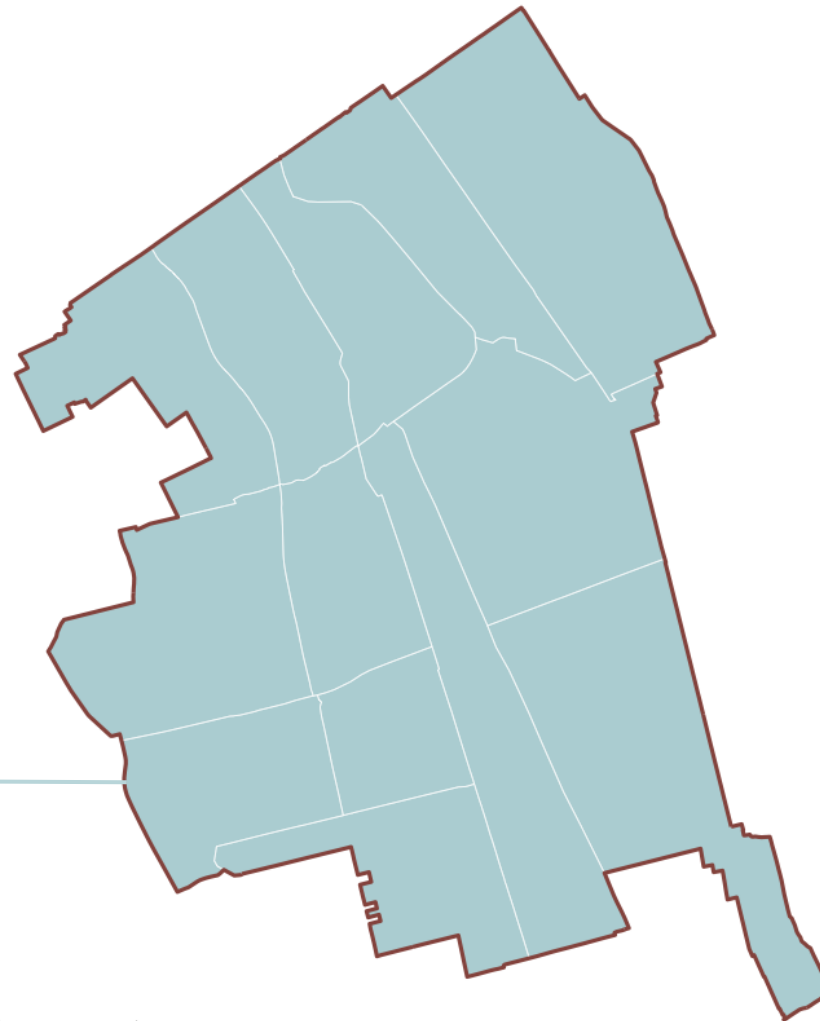
population of 102.000

24,4 KM2

31 % green coverage

low density

historical core



1 km



20 km

CITY PAIR

YUEXIU DISTRICT

population of 1.173.000

33,8 KM2

12 % green coverage

high density

historical core



DELFT

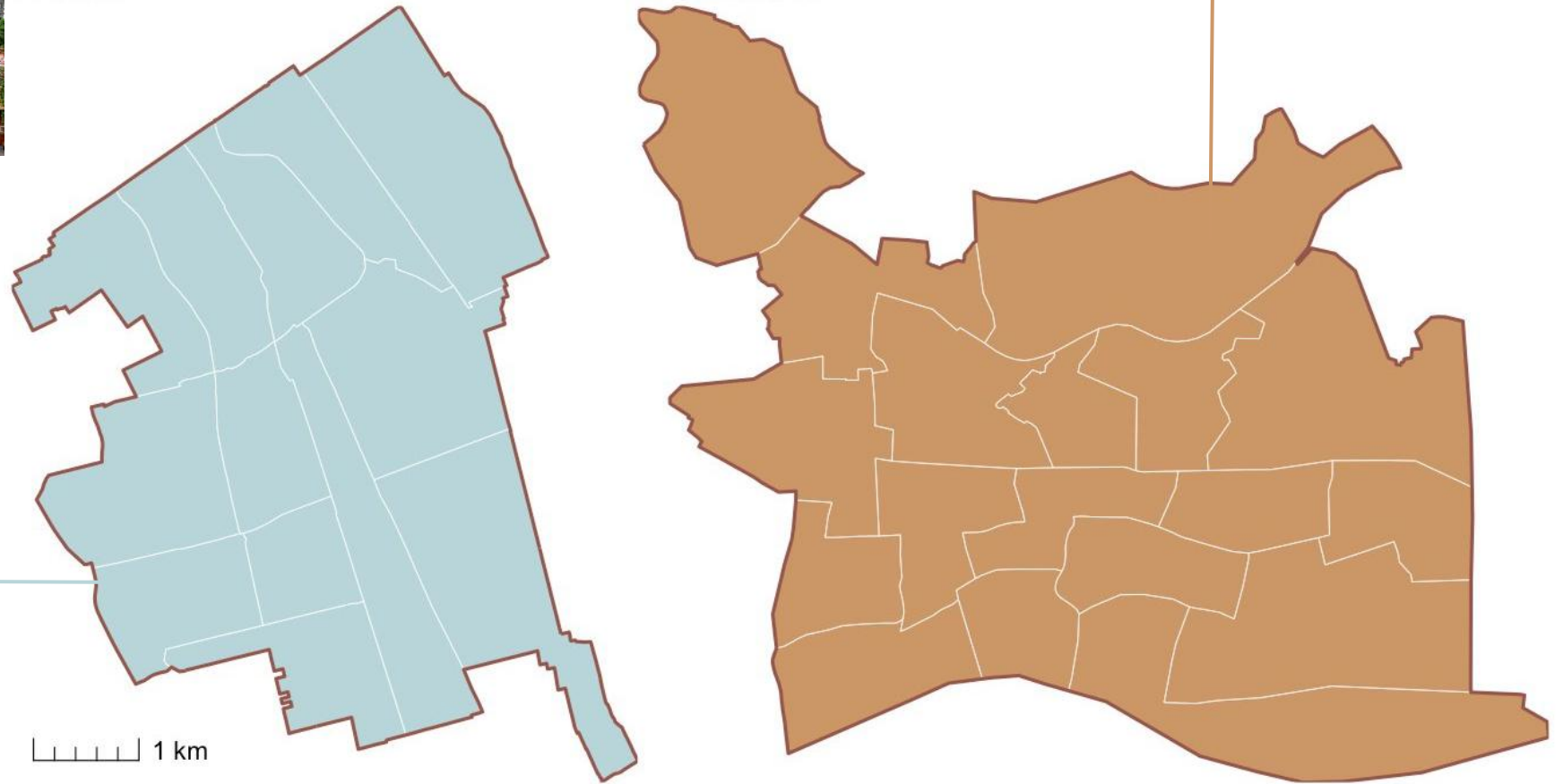
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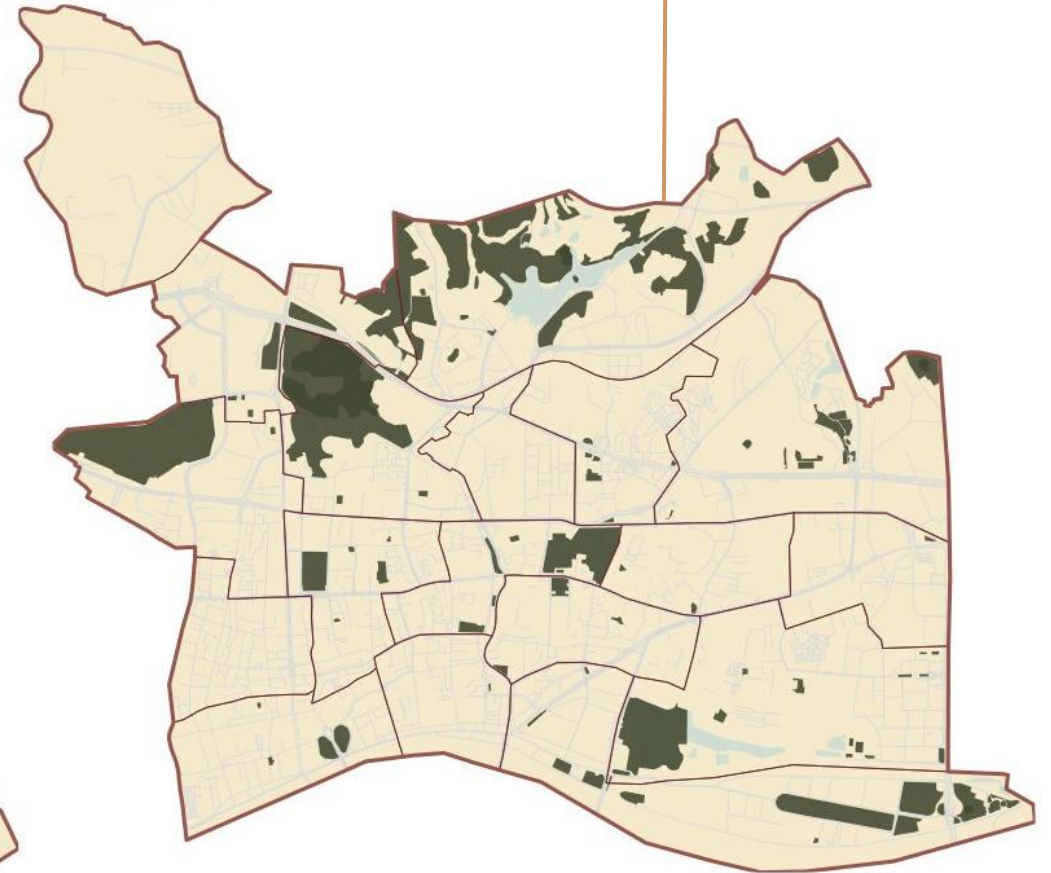
population of 102.000

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URBAN GREEN SPACE

How does **connectivity** between & **biodiversity** in **urban green spaces** contribute to **spatial justice** in the cities Delft and Guangzhou?

What are opportunities for improvement?

RESEARCH QUESTION

SQ1 How accessible are green spaces?

SQ2 What are the different typologies and their ecological value?

SQ3 How equitable are green spaces distributed?

SQ4 How do Delft and Guangzhou perform?

SQ5 What NBS can we suggest to improve spatial justice with urban green space connectivity?

Accessibility

Biodiversity & typology

Connectivity

Performance (MCDA)

Suggestions

RESEARCH SUB-QUESTIONS

ACCESSIBILITY

population density
green space per capita
buffer analysis
mean nearest distance

BIODIVERSITY

green space typology
NDVI vegetation density
species observation
blue-green ratio

SPATIAL JUSTICE

green space Gini-coefficient
bivariate choropleth
green space & income correlation

CONNECTIVITY

fragmentation indices
functional connectivity graph
NBS corridor prioritization

MCDA

accessibility 0.3
biodiversity 0.25
spatial justice 0.25
connectivity 0.20

METHODOLOGY

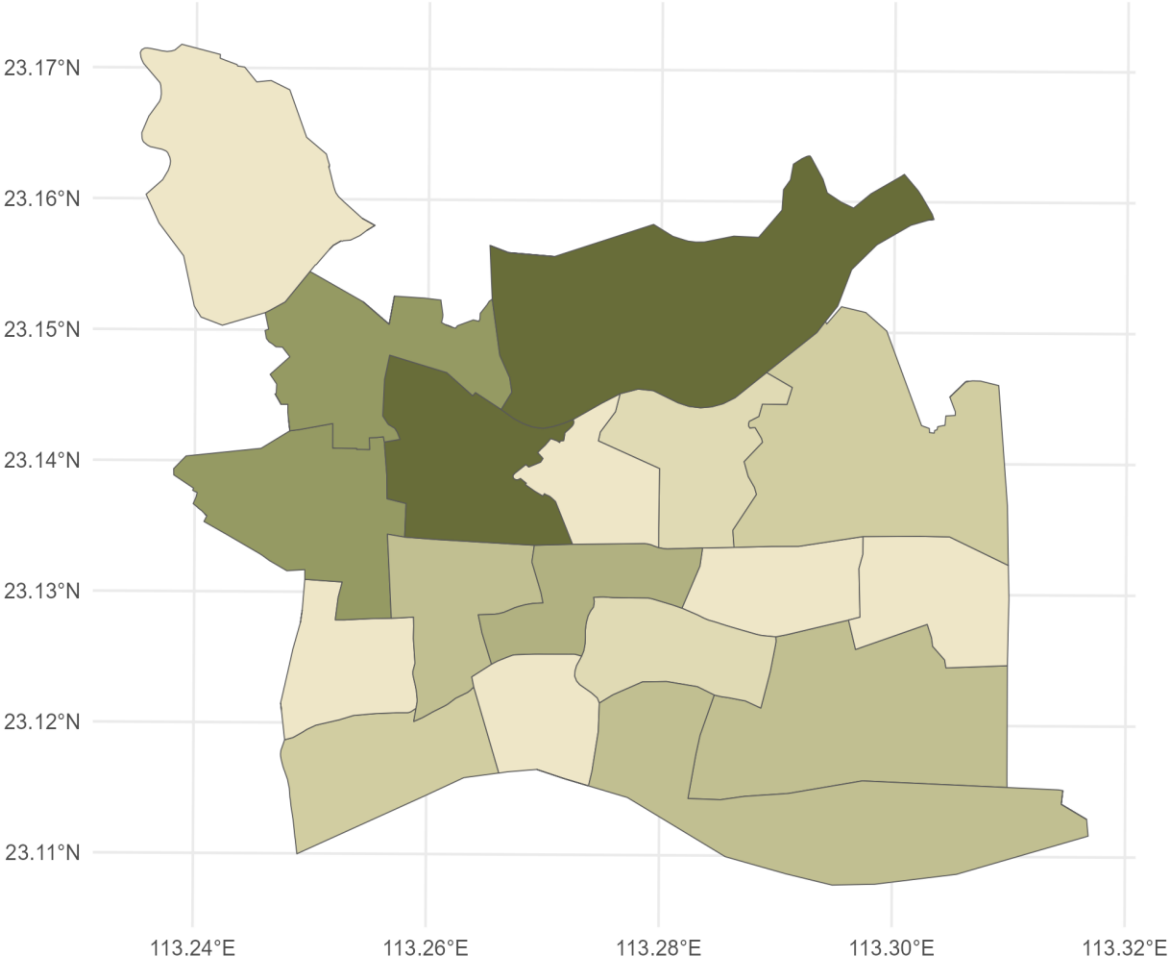
Metric	Yuexiu	Delft
Median green per person	0.07 m ²	25 m ²
Meets WHO minimum (9m ²)?	0 of 18 districts	13 of 13
Walk to nearest green	208m median	56m median

ACCESSIBILITY

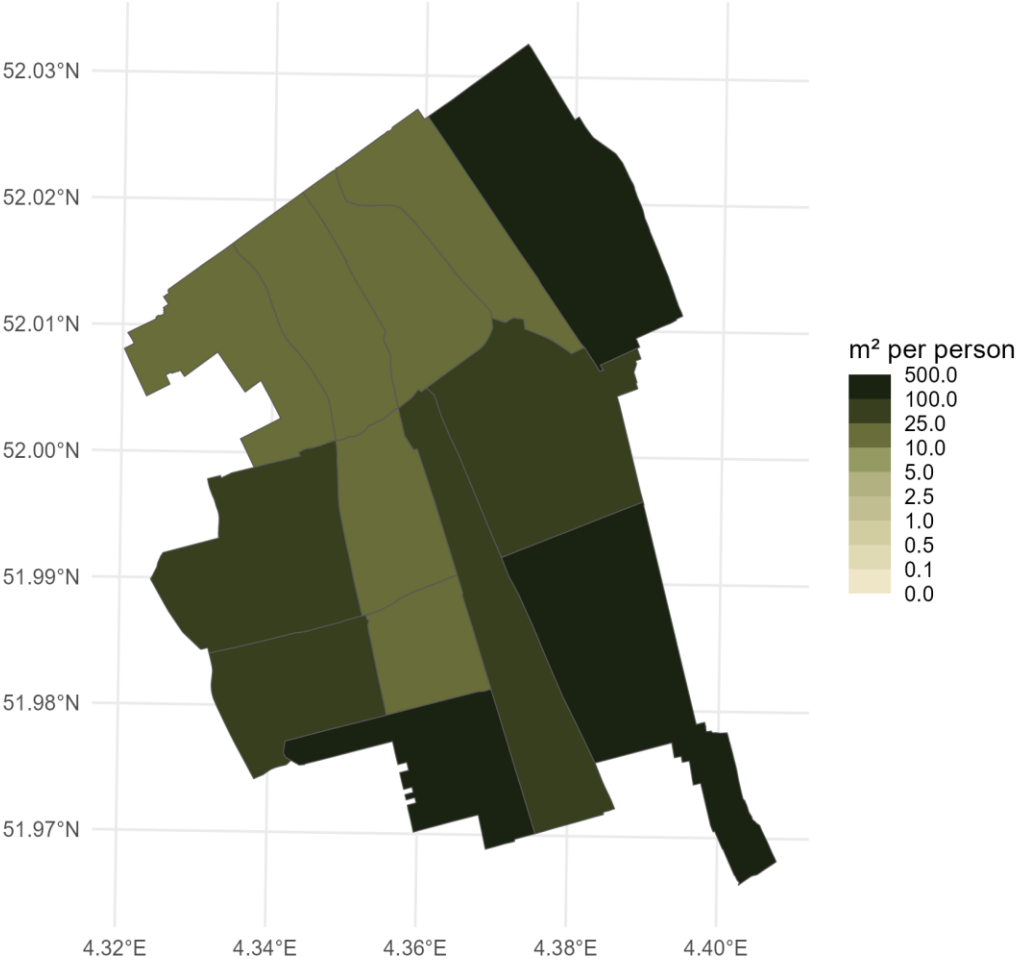
*Can residents actually reach
green space?*

ACCESSIBILITY

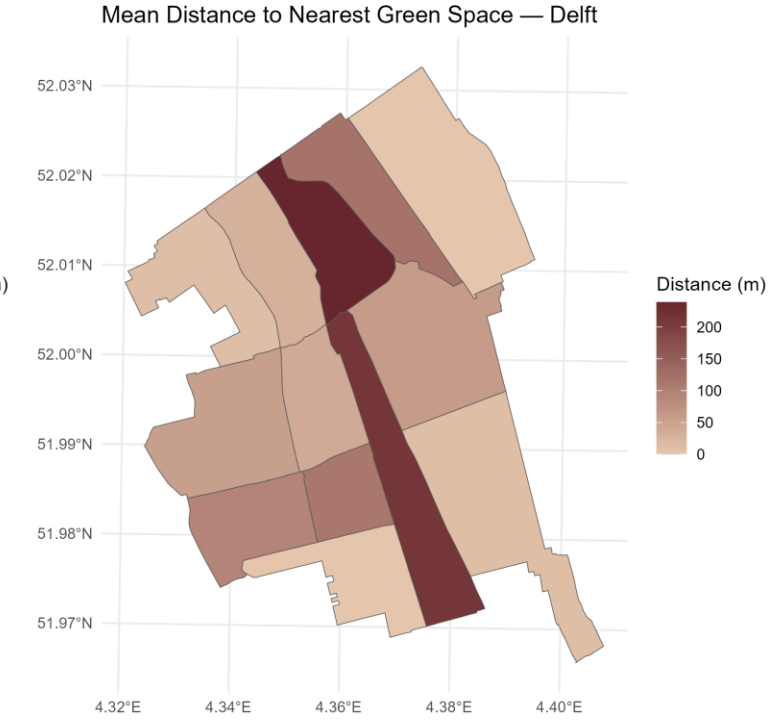
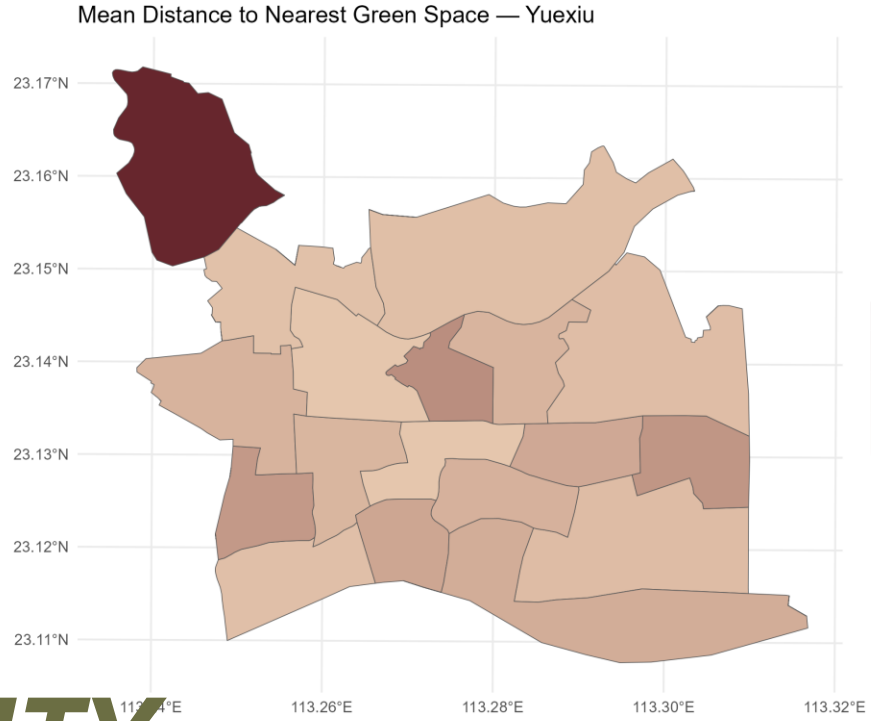
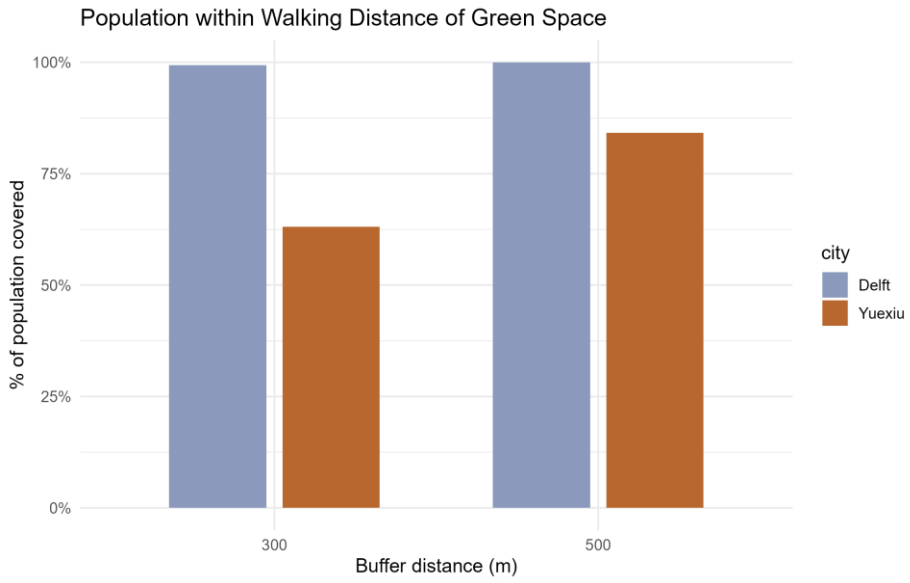
Green Space per Capita — Yuexiu Jiědào



Green Space per Capita — Delft Wijken



ACCESSIBILITY



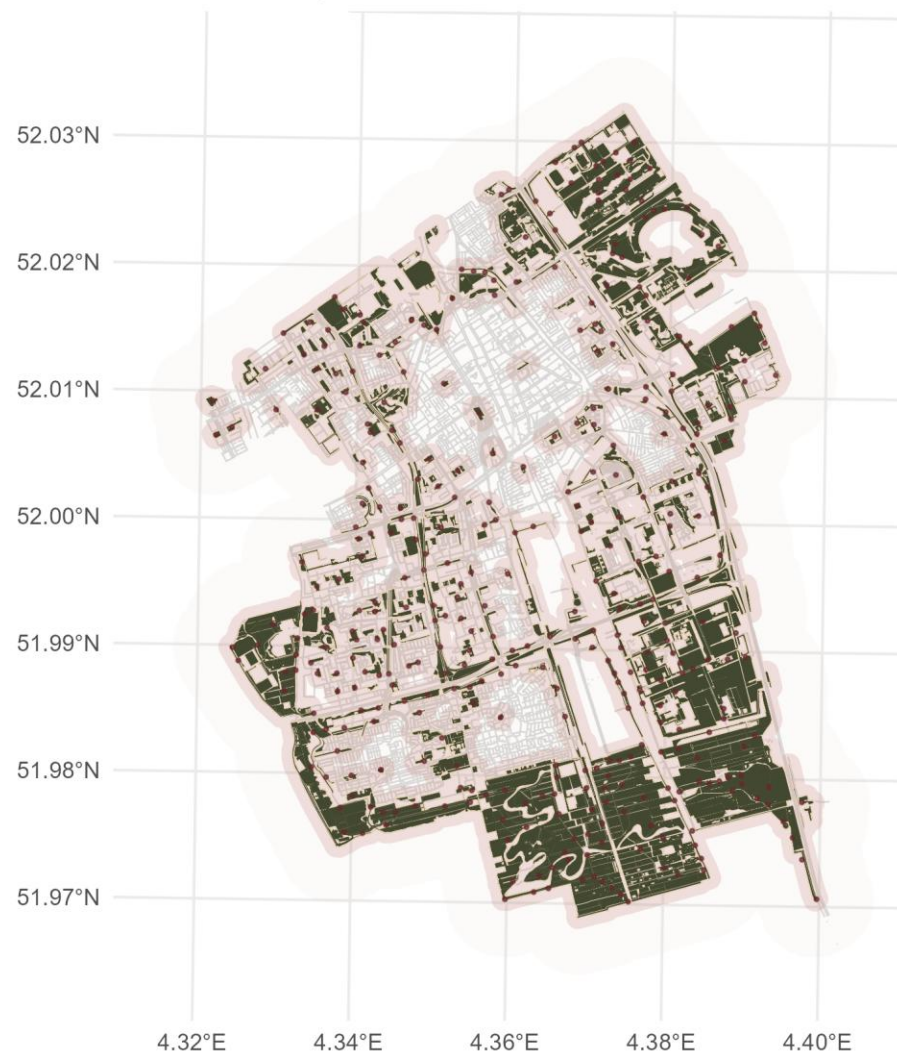
ACCESSIBILITY

ACCESS POINTS TO GREEN SPACE

Network Entry Thresholds — Yuexiu



Network Entry Thresholds — Delft



ACCESSIBILITY

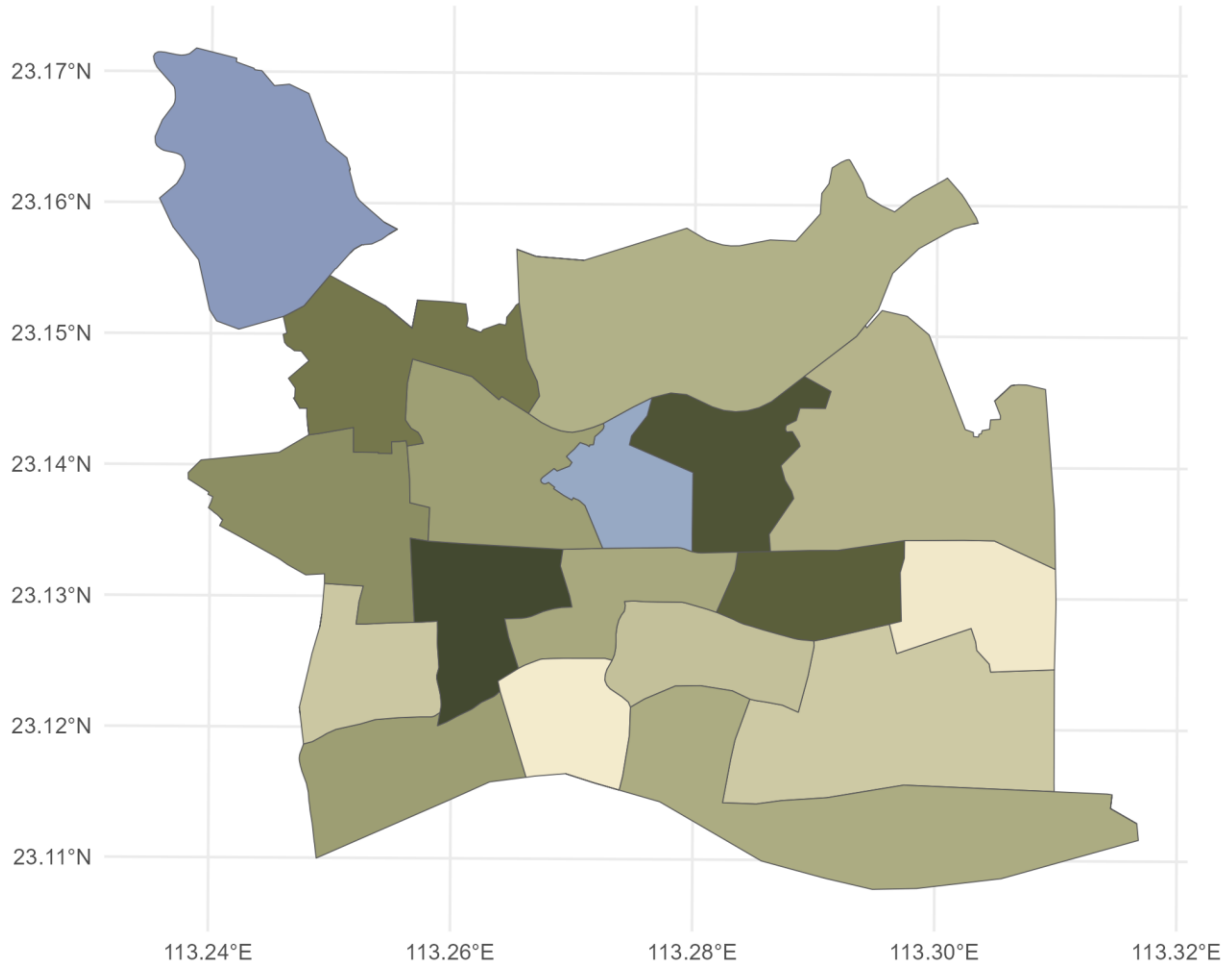
Metrics	Yuexiu	Delft
Number of green patches	114	4,039
Mean patch size	3.5 ha	0.46 ha
Green type	65% formal parks	73% semi-natural

BIODIVERSITY

What kind of green exists, and how healthy is it?

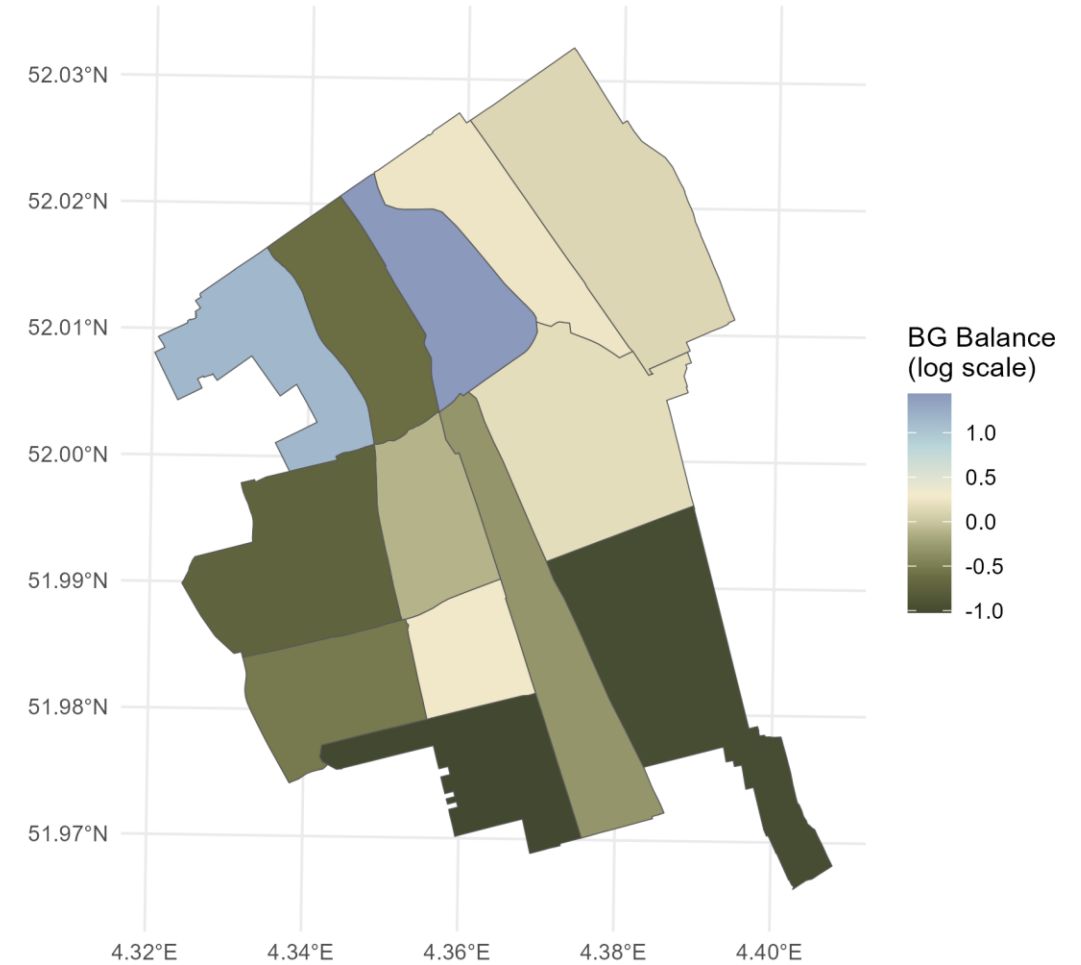
Blue-Green Balance — Yuexiu Subdistricts

$\log_{10}(\text{Water PC}) - \log_{10}(\text{Green PC})$



Blue-Green Balance — Delft Wijken

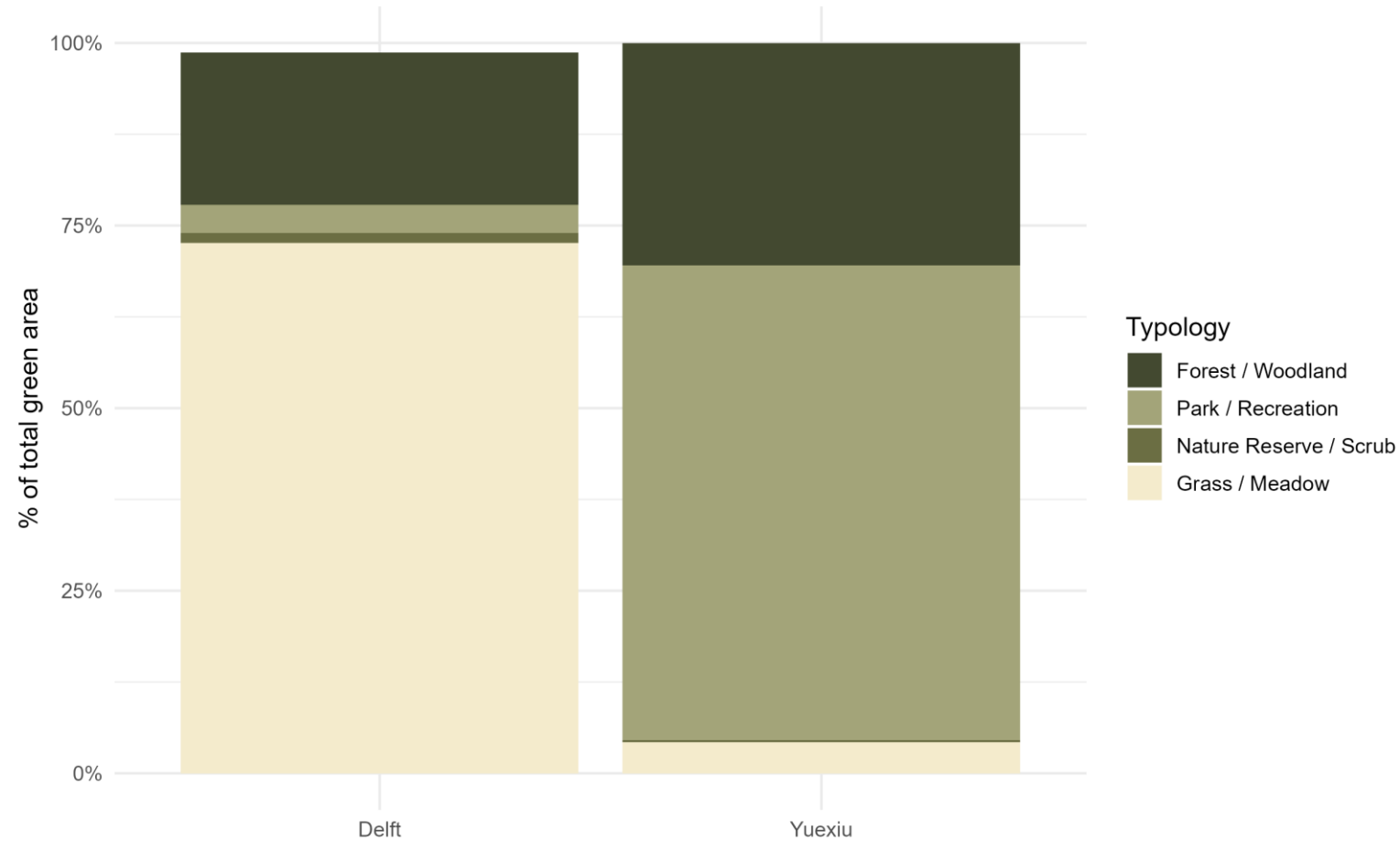
$\log_{10}(\text{Water PC}) - \log_{10}(\text{Green PC})$



BIODIVERSITY

Both cities exhibit a balanced blue-green ratio, reflecting efforts on urban cooling and biodiversity

Green Space Composition by Typology
OSM/BGT tag-based classification — % of total green area

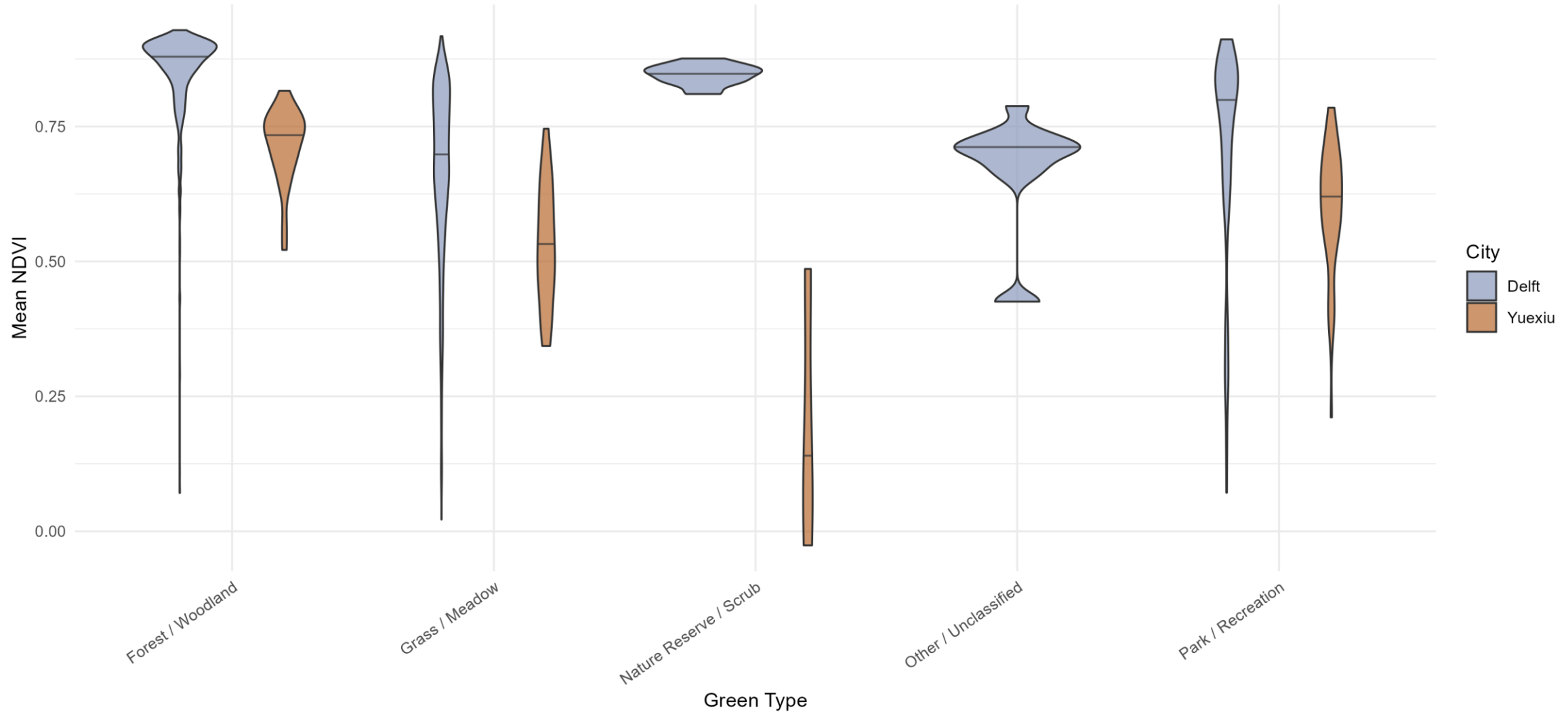


Delft's green space is dominated by meadows/grassland, consistent with its character as a smaller European city.

Yuexiu is dominated by formal parks/recreation areas, reflecting the requirements of a dense, historic urban center.

NDVI Distribution by Green Typology

Patch-level mean NDVI (vegetation health proxy)



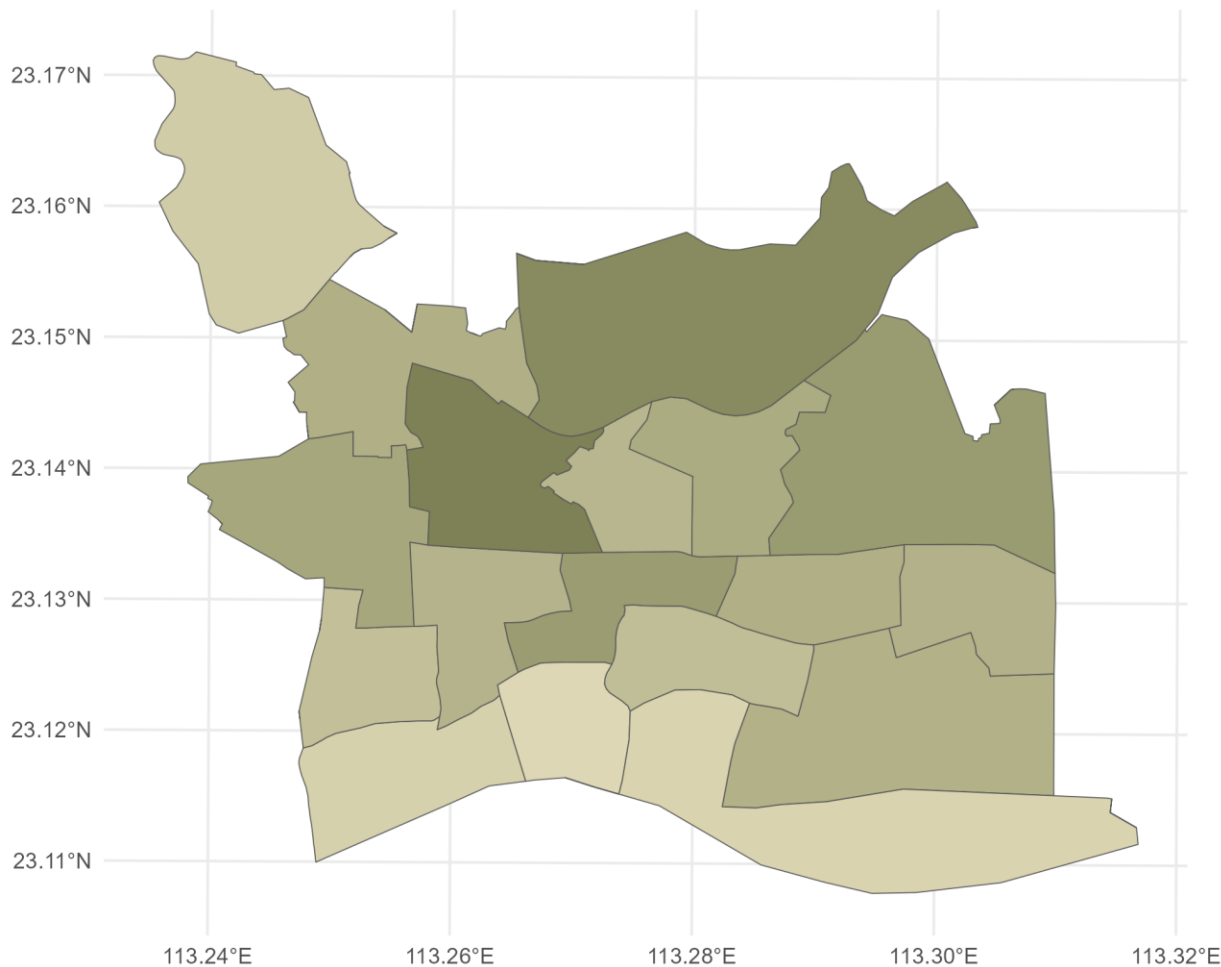
BIODIVERSITY

Between-city interpretation

- Delft's tight cluster → indicate high quality.
- Yuexiu's violin display wider spread → high variability in green space quality

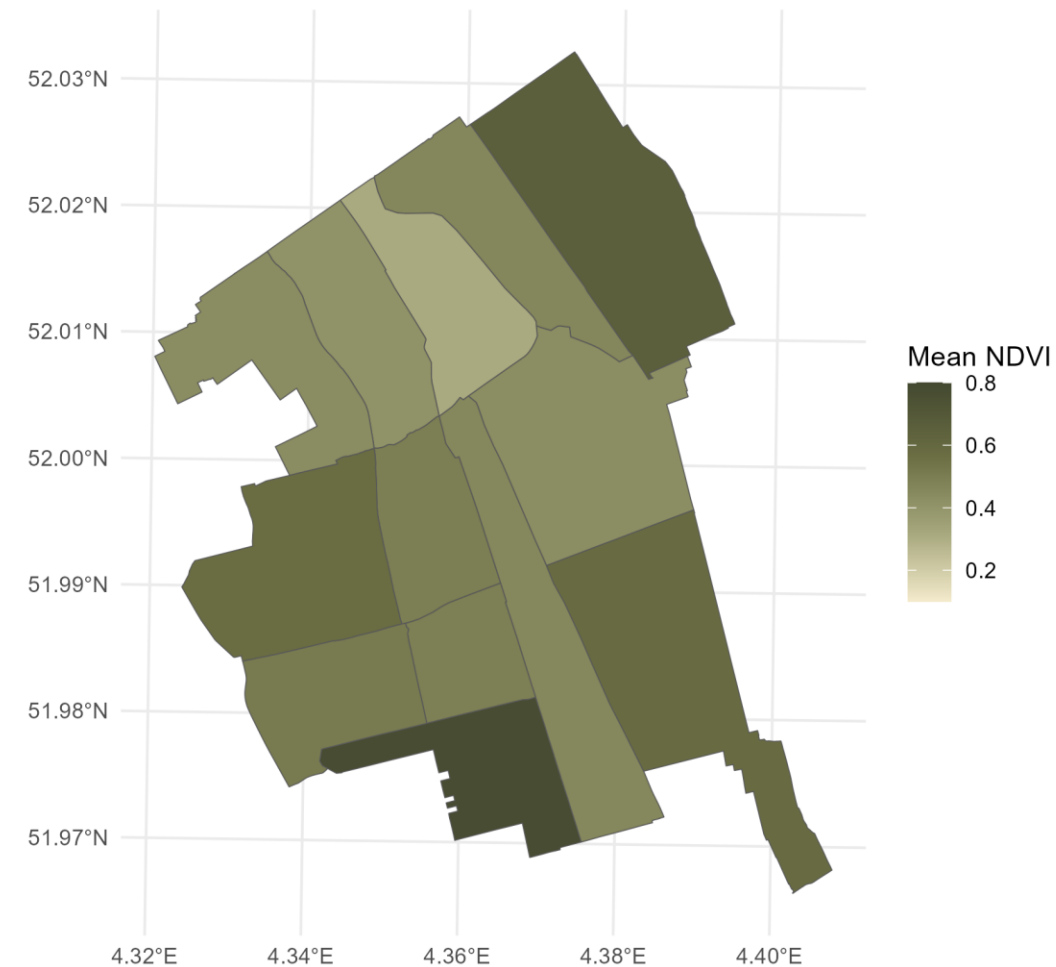
Mean NDVI per Subdistrict — Yuexiu

Source: Sentinel-2 / Landsat



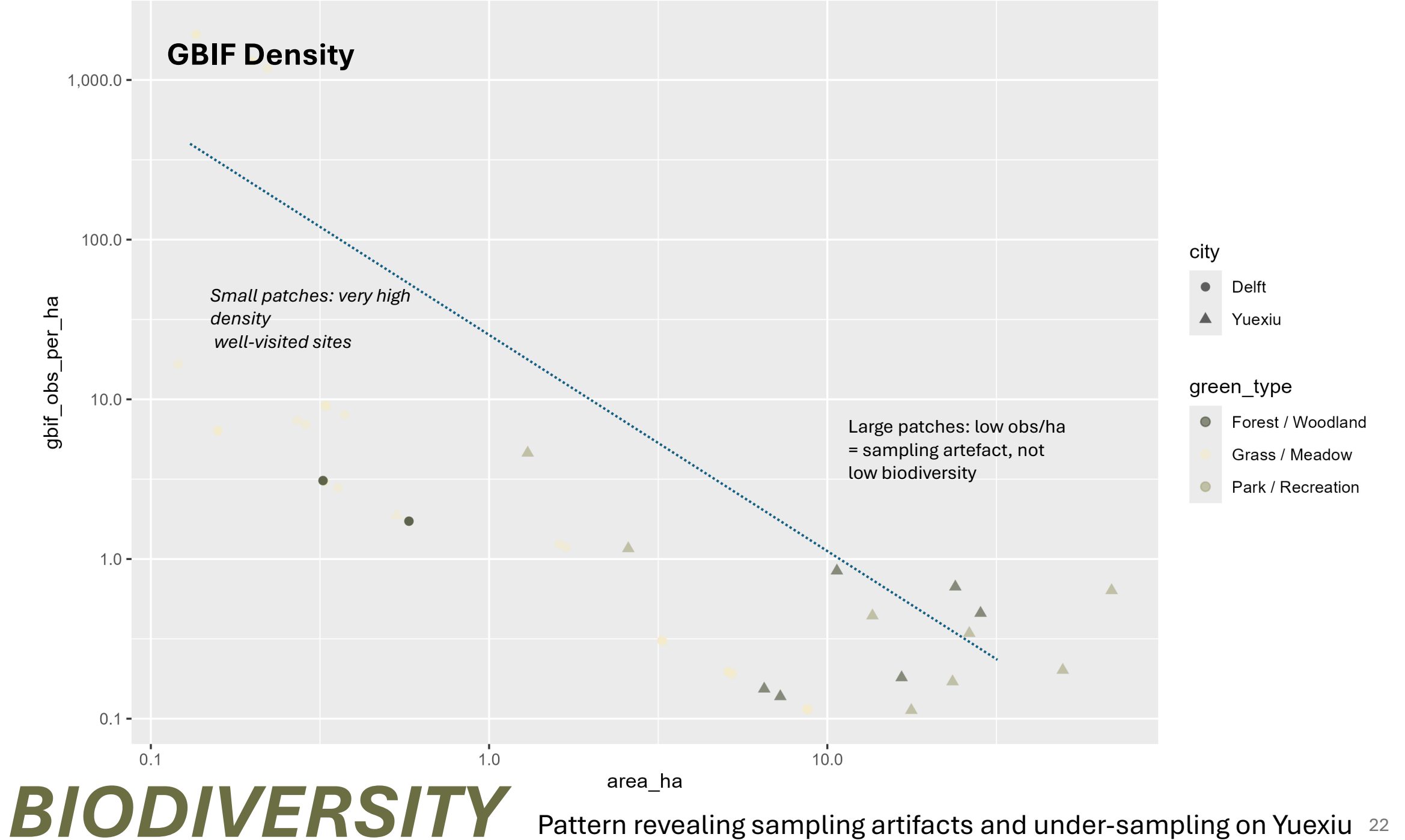
Mean NDVI per Wijk — Delft

Source: Sentinel-2 / Landsat



Regardless of macro-level spatial distribution - Delft has denser & higher-quality vegetation, whereas Yuexiu has more sparse greenery

BIODIVERSITY



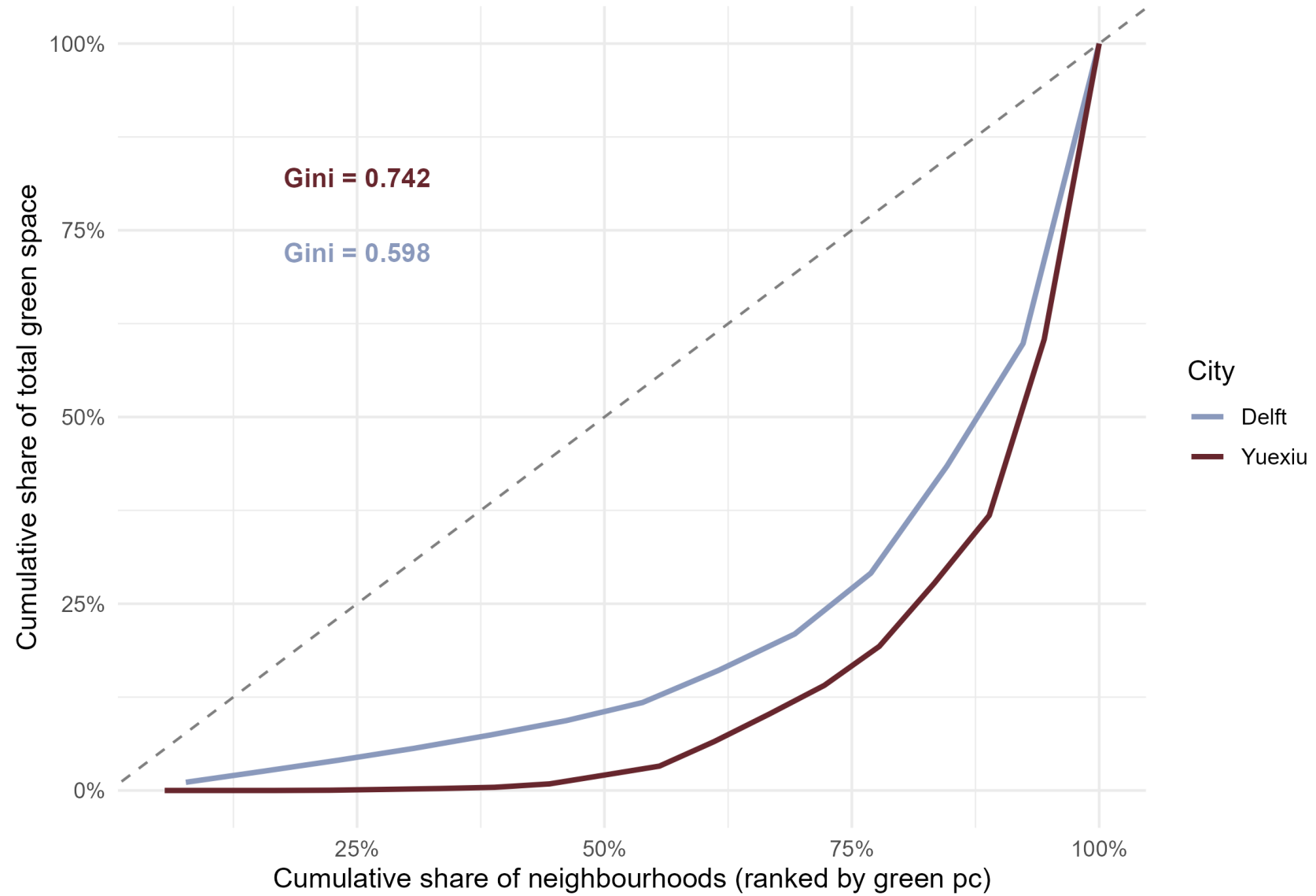
Metrics	Yuexiu	Delft
Gini coefficient	0.742	0.598
Bottom 45% of districts hold	~0% of total green	~15% of total green
Green-population mismatch	Spatially concentrated south	Concentrated in Binnenstad

SPATIAL JUSTICE

Is green space distributed fairly
across the population?

Lorenz Curve — Green Space per Capita Inequality

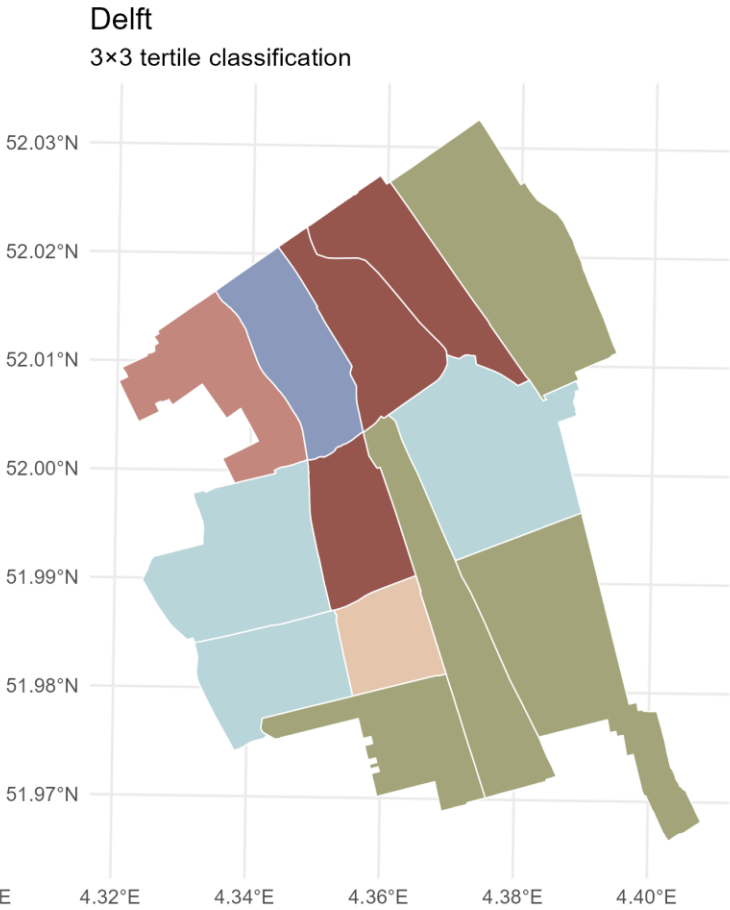
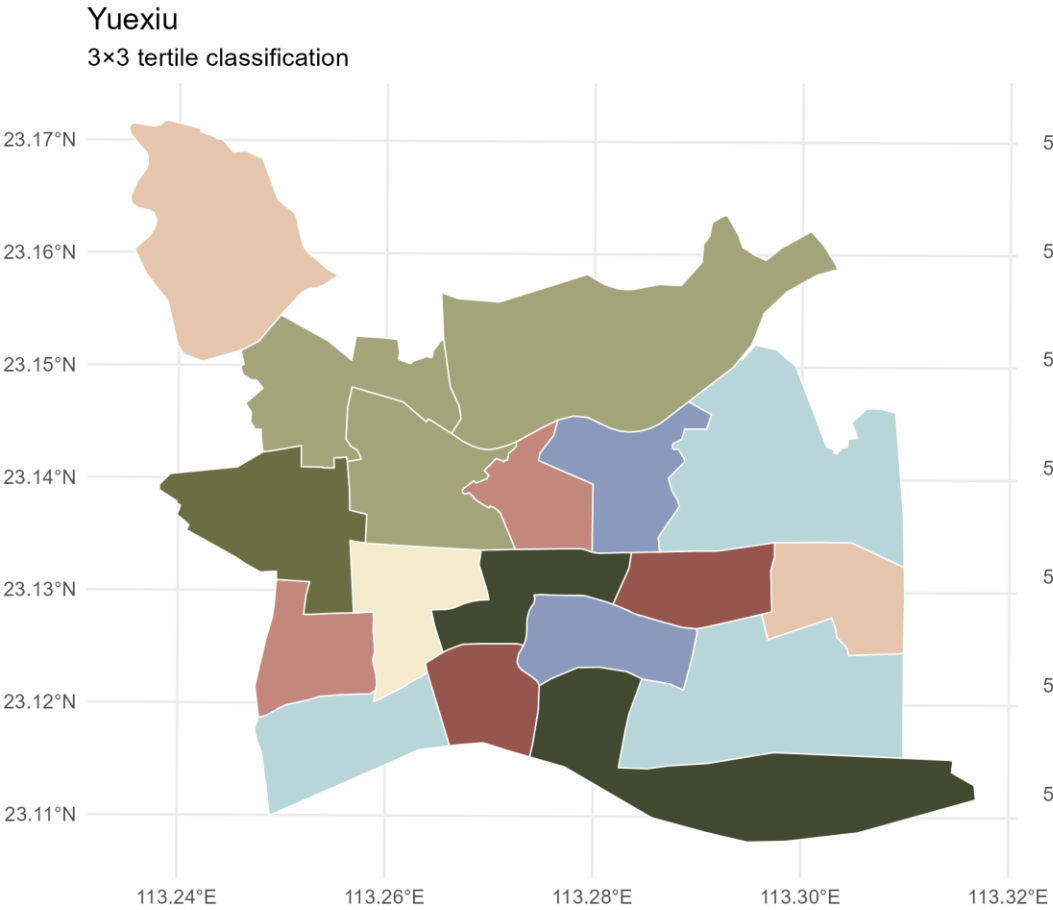
Diagonal = perfect equality; further below = more unequal



SPATIAL JUSTICE

BIVARIATE: GREEN DENSITY X POPULATION DENSITY

Bivariate: Green Density × Population Density



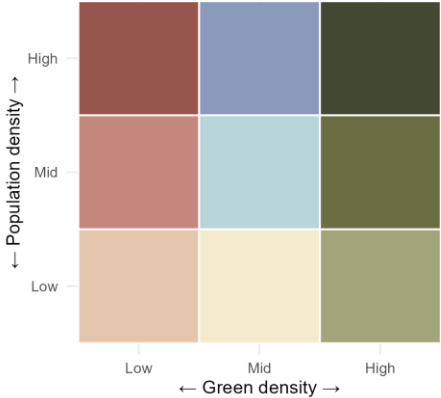
Green–Population matrix

- Low-Low
- Low-Mid
- Low-High
- Mid-Low
- Mid-Mid
- Mid-High
- High-Low
- High-Mid
- High-High

Green–Population matrix

- Low-Low
- Low-Mid
- Low-High
- Mid-Low
- Mid-Mid
- Mid-High
- High-Low
- High-Mid
- High-High

Bivariate legend

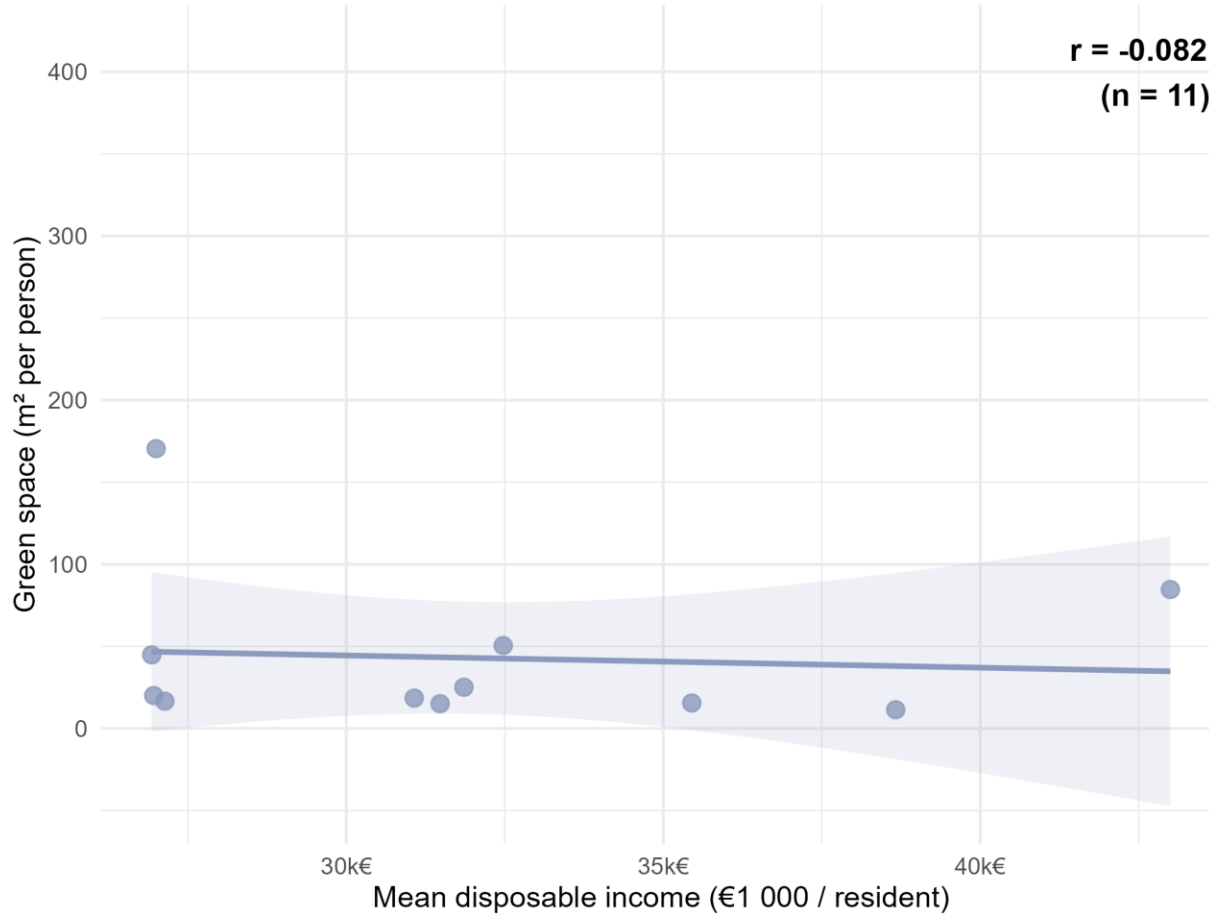


SPATIAL JUSTICE

Equity Correlations

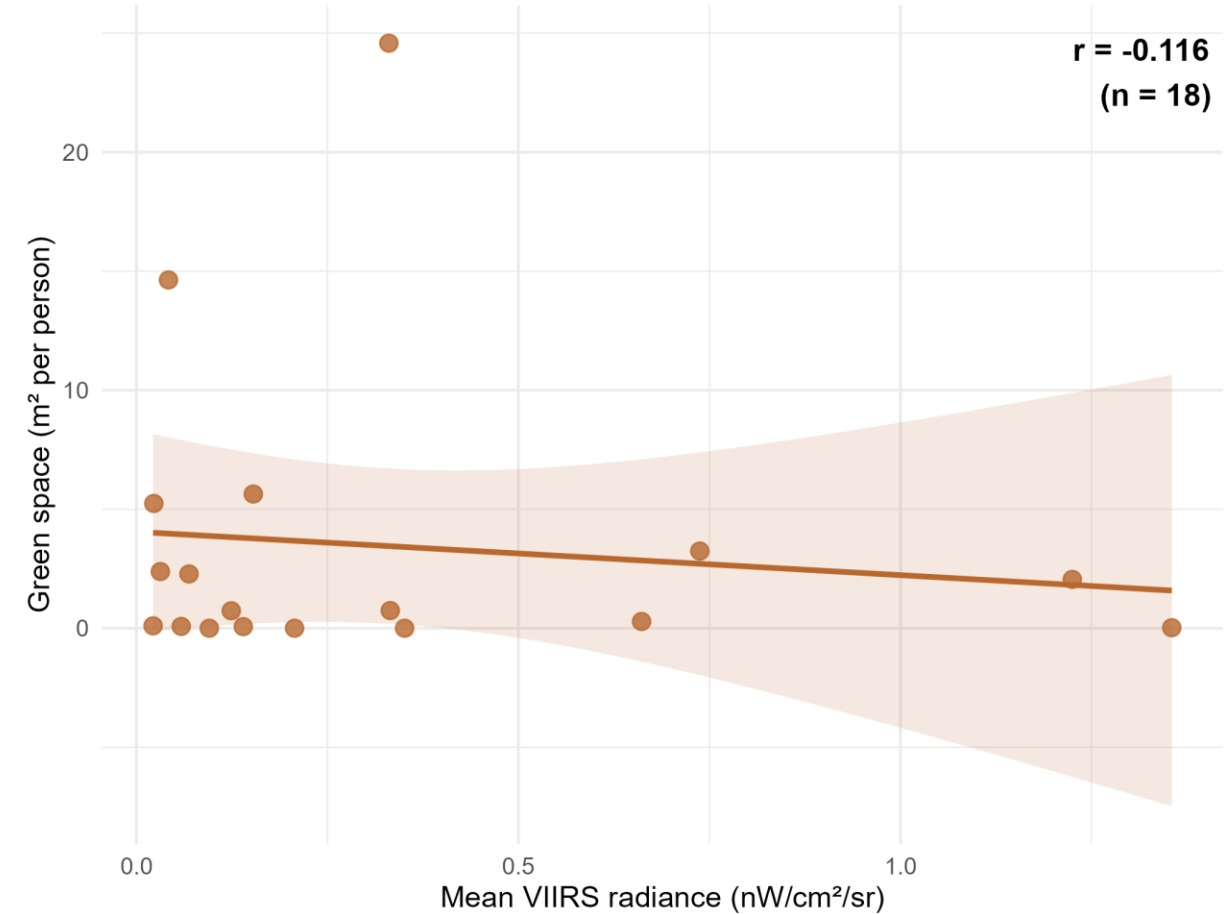
Green Space per Capita vs. Household Income — Delft

Pearson correlation; wijk level (residential wijken only)



Green Space per Capita vs. VIIRS Nighttime Light — Yuexiu

Pearson correlation; subdistrict level (residential only)



SPATIAL JUSTICE

Metrics	Yuexiu	Delft
Dispersal edges	42	37,936
Isolated components	83	90
Largest connected cluster	7 patches	1,356 patches

CONNECTIVITY

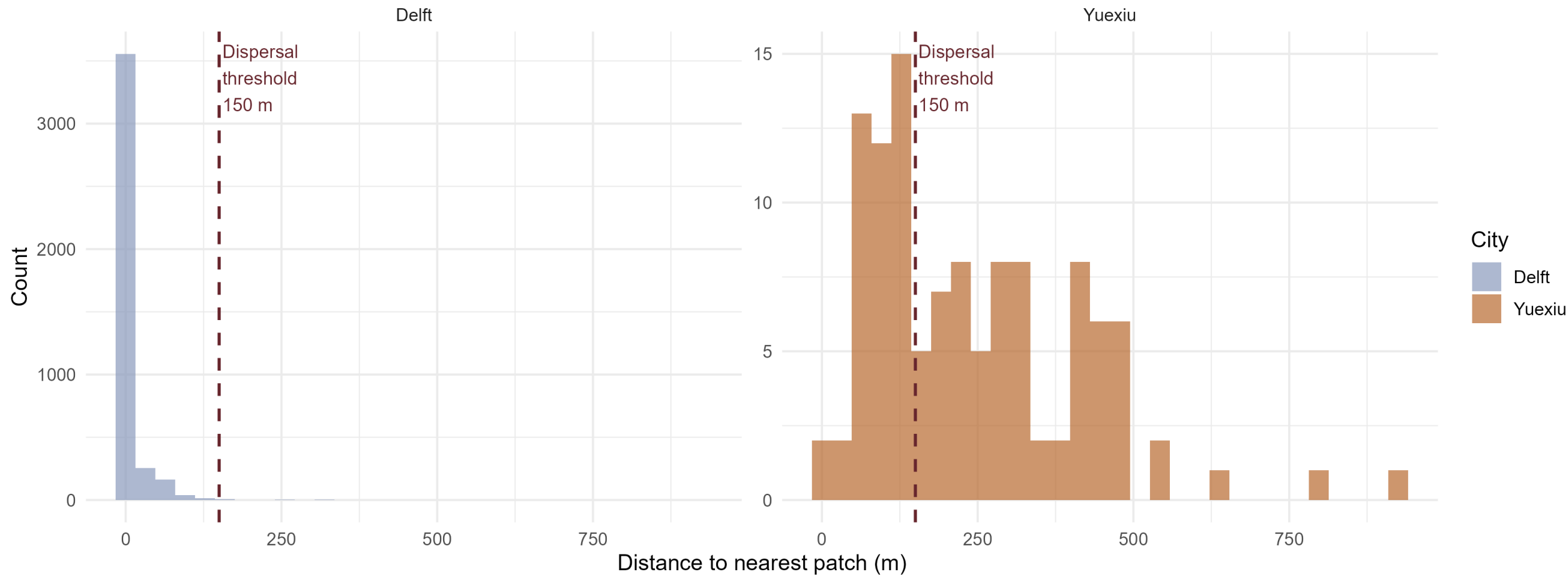
How connected are these green spaces?

CONNECTIVITY

EUCLIDEAN NEAREST-NEIGHBOUR DISTANCE BETWEEN GREEN PATCHES

Euclidean Nearest-Neighbour Distance Between Green Patches

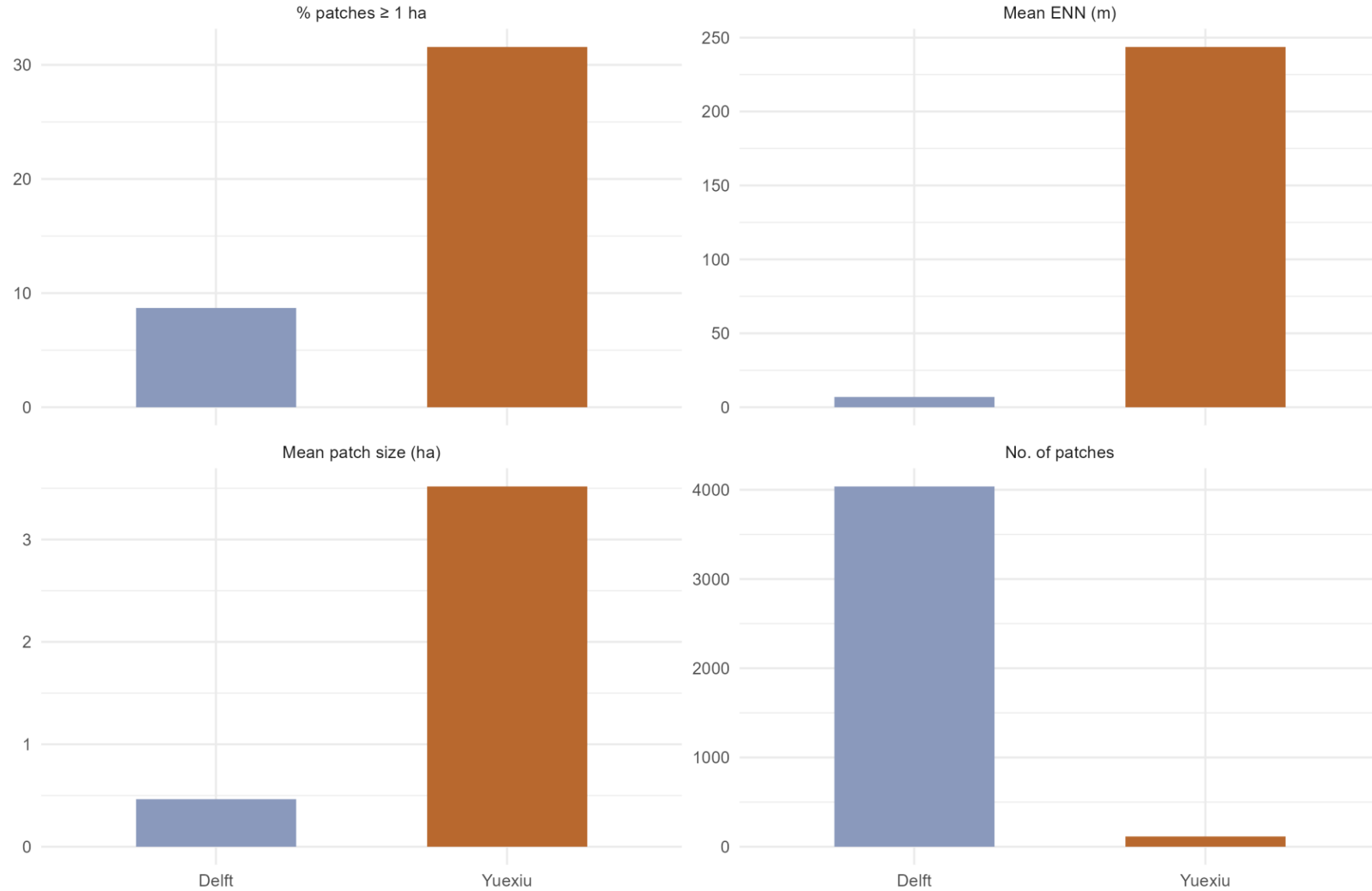
Dashed line = dispersal threshold for connectivity graph



CONNECTIVITY

Landscape Fragmentation Metrics

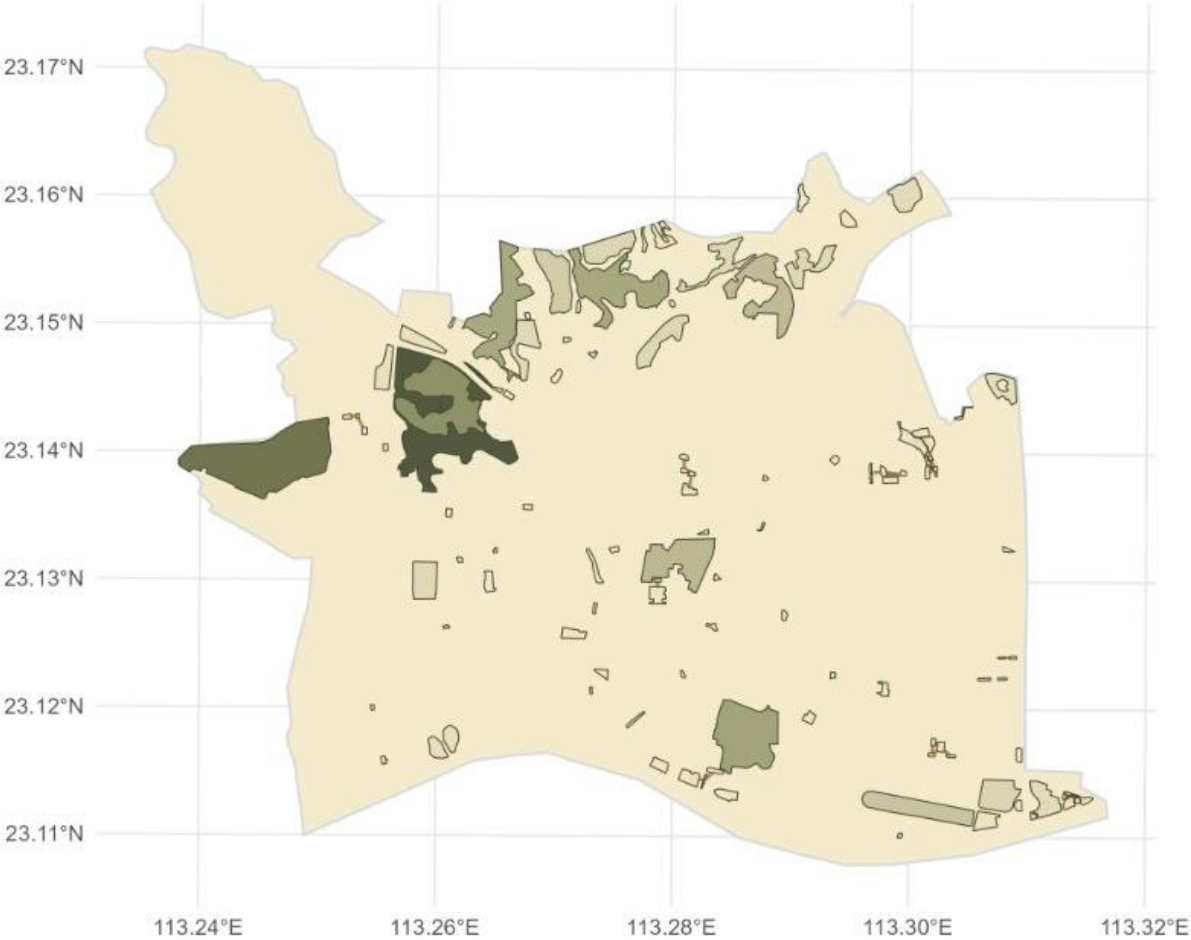
Computed from OSM green patch vectors



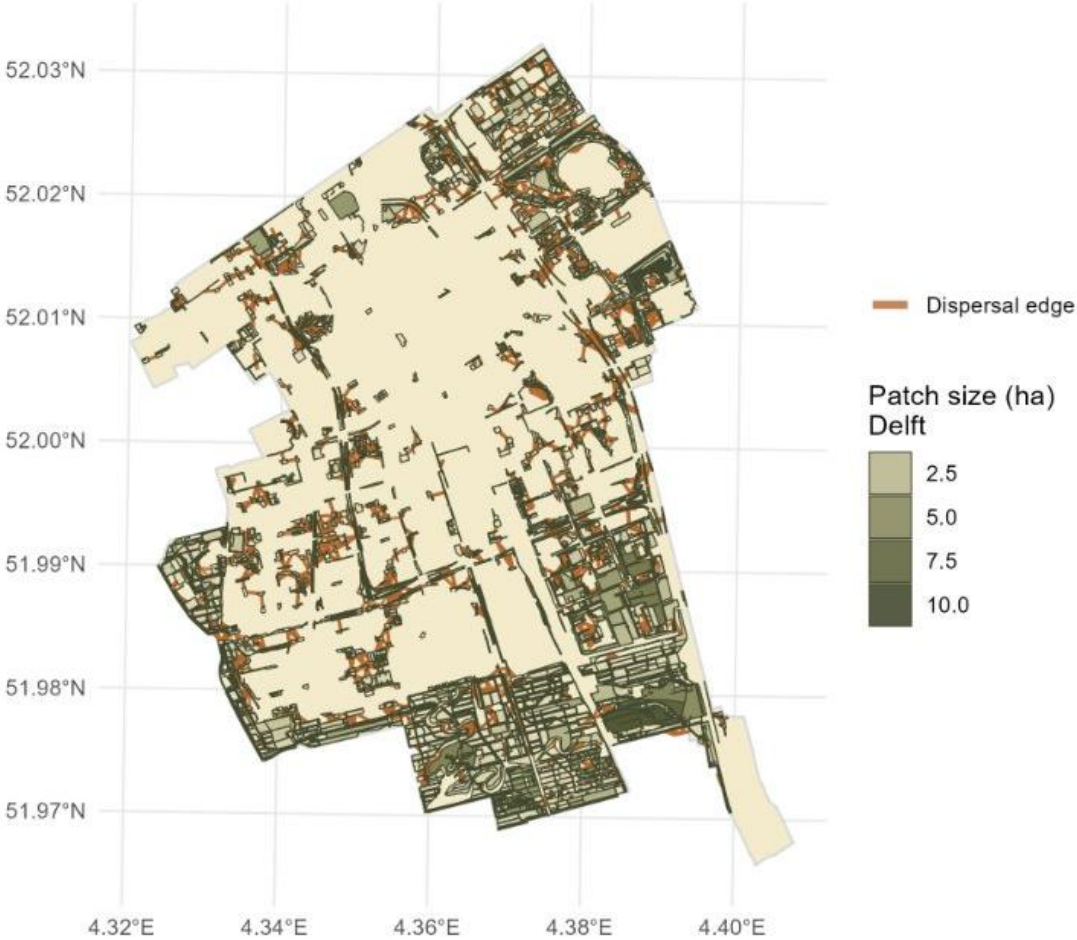
CONNECTIVITY

FUNCTIONAL CONNECTIVITY GRAPH

Green Patch Network — Yuexiu
Dispersal threshold: 150 m | Nodes sized by patch area



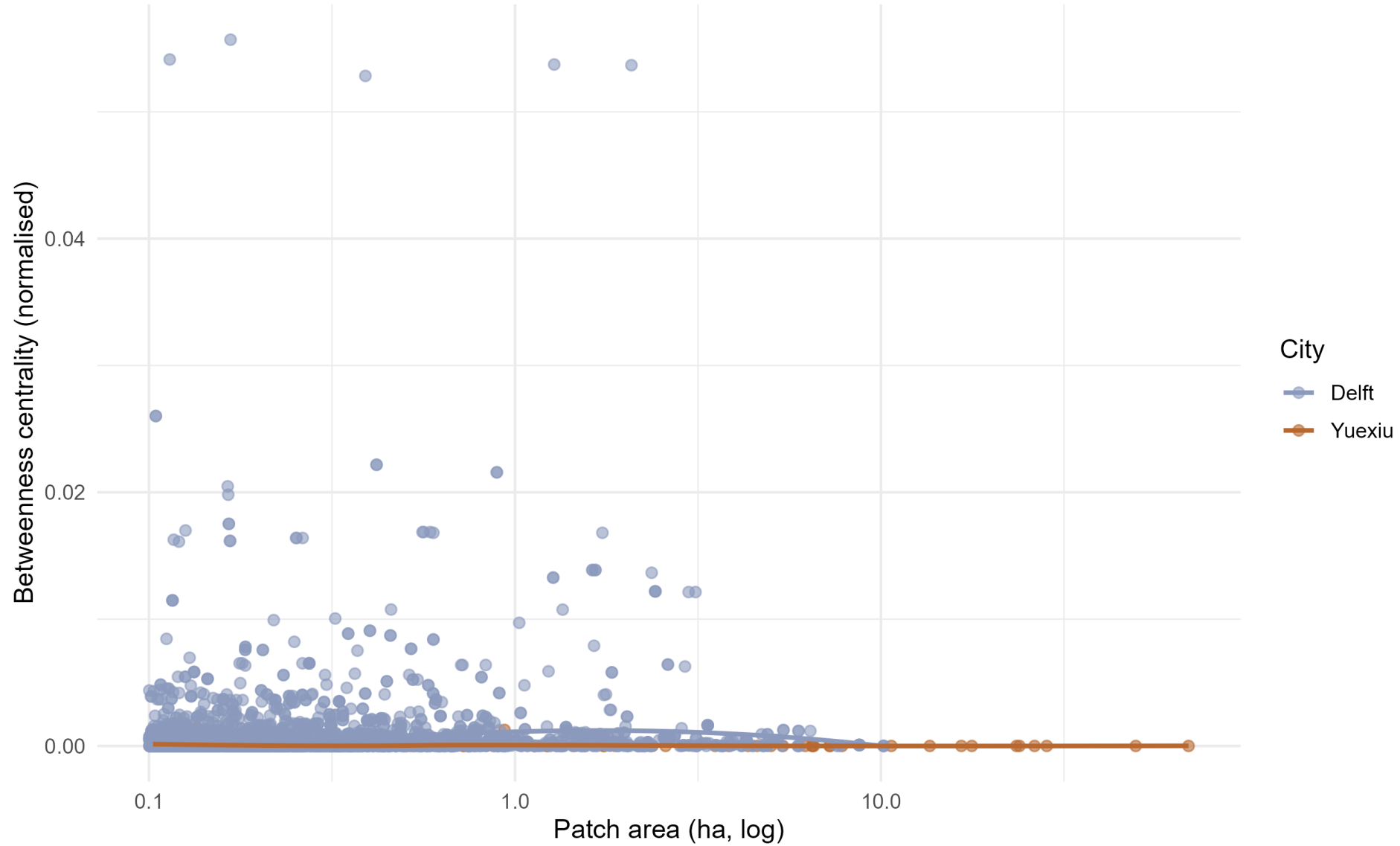
Green Patch Network — Delft
Dispersal threshold: 150 m | Nodes sized by patch area



CONNECTIVITY

Patch Area vs. Betweenness Centrality

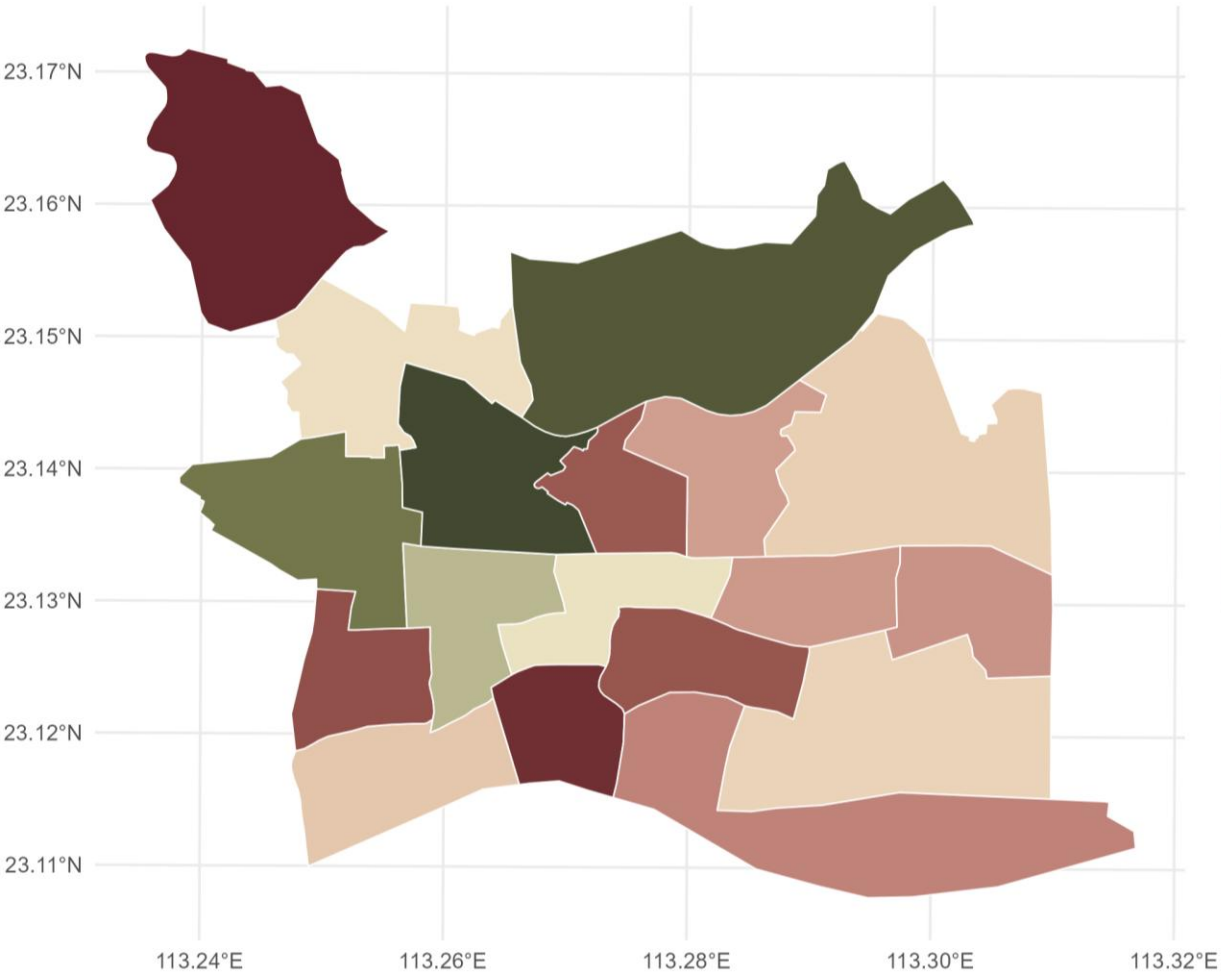
High betweenness = stepping-stone patch critical for connectivity



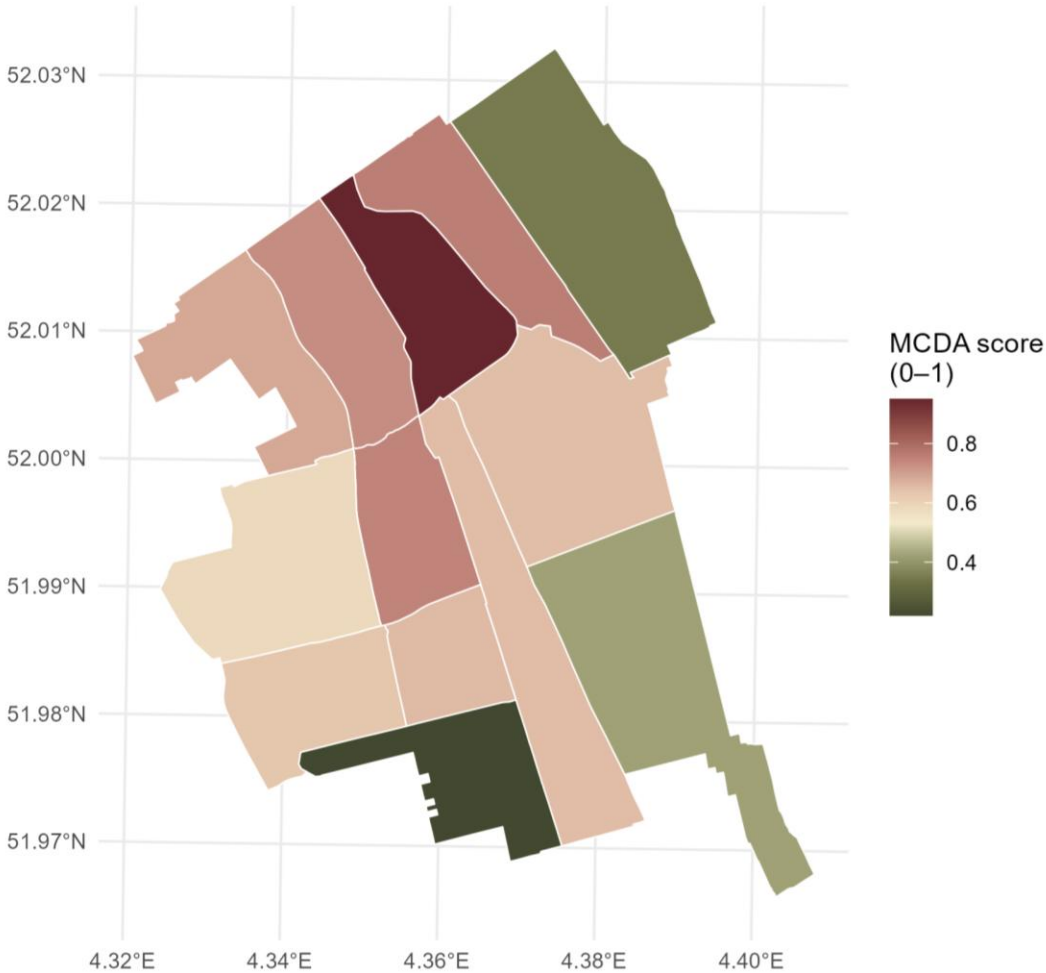
Metrics	Yuexiu	Delft
High priority zones	6 of 18 subdistricts	4 of 13 wijken
Highest scoring unit	Kuangquan — 0.832	Binnenstad — 0.95
Primary driver	Connectivity + biodiversity	Access + biodiversity
What's needed	Build a network from scratch	Protect and fill gaps

MCDA COMPOSITE SCORE

Weights: Acc 30% | Bio 25% | Conn 25% | Eq 20%

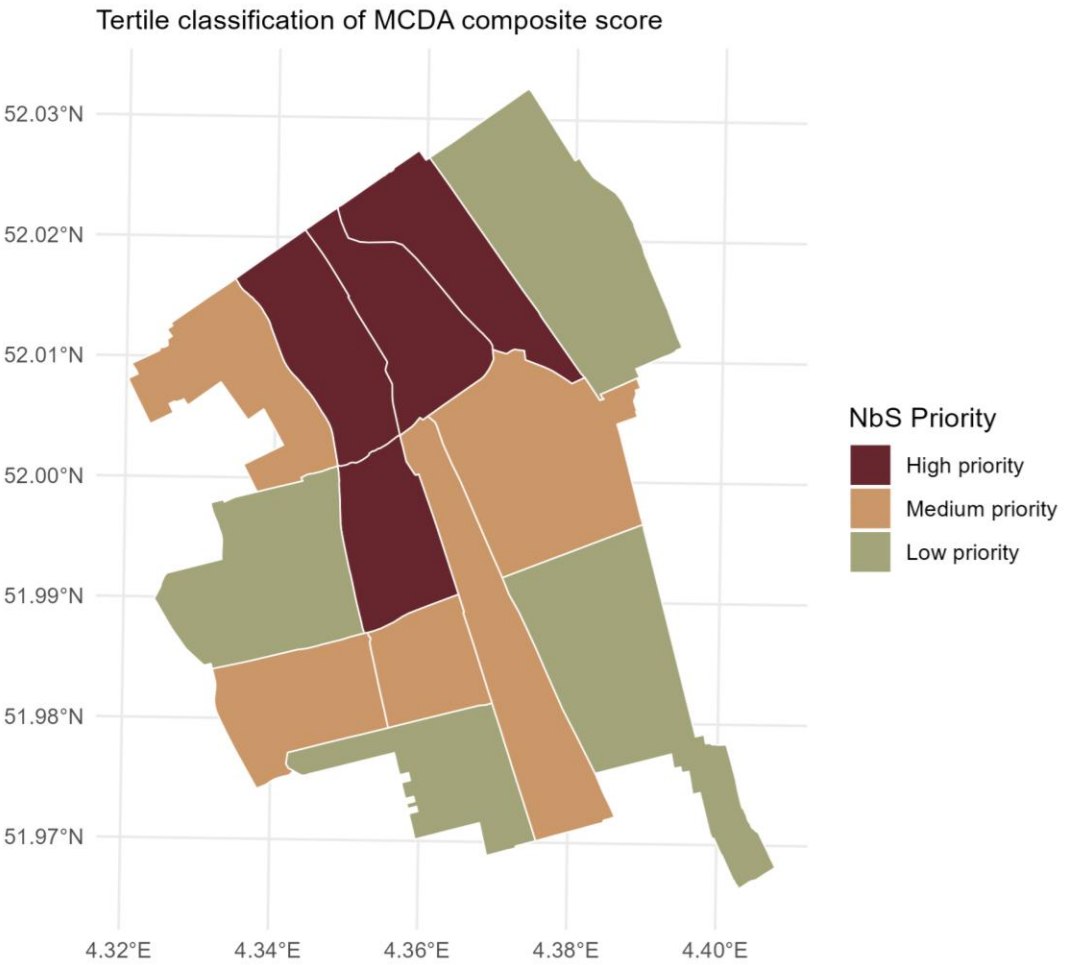
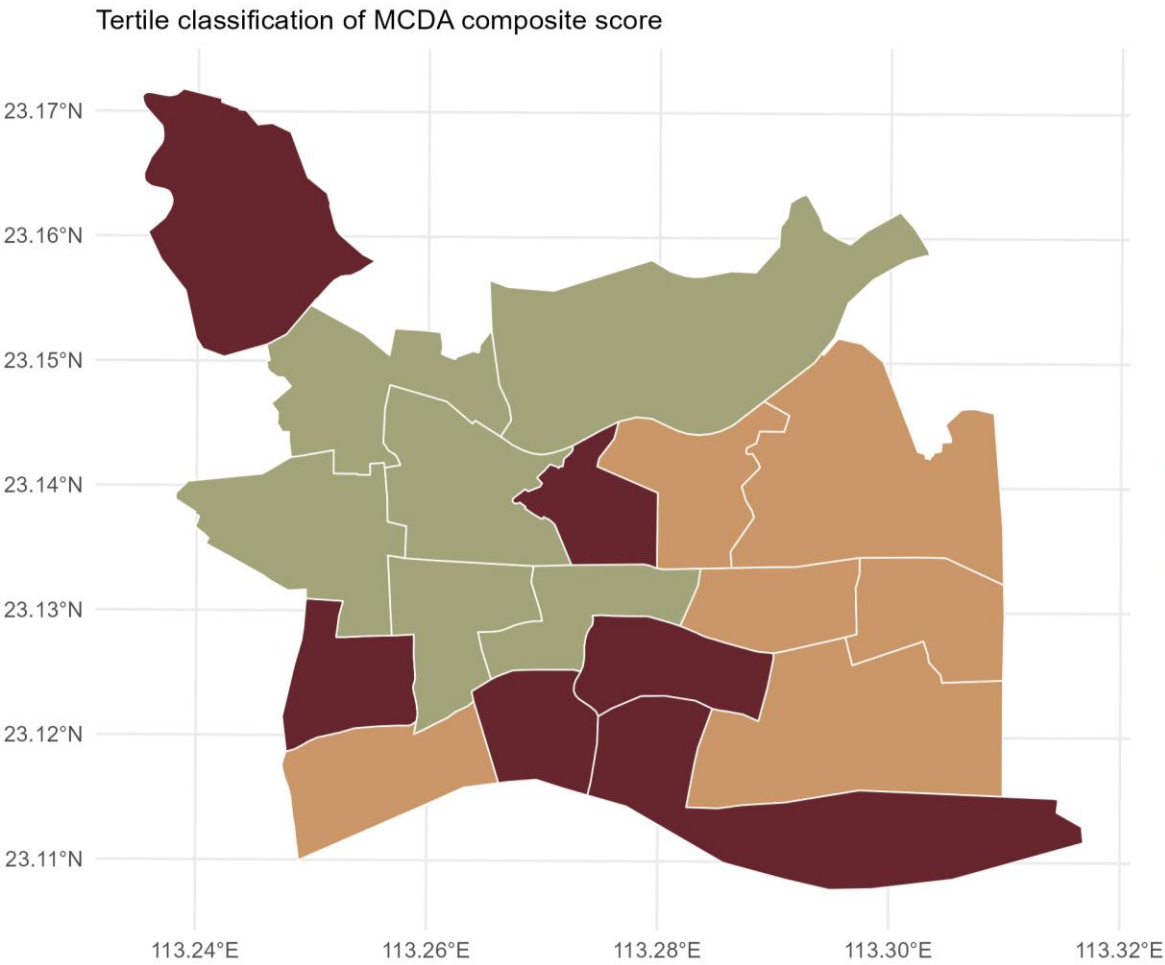


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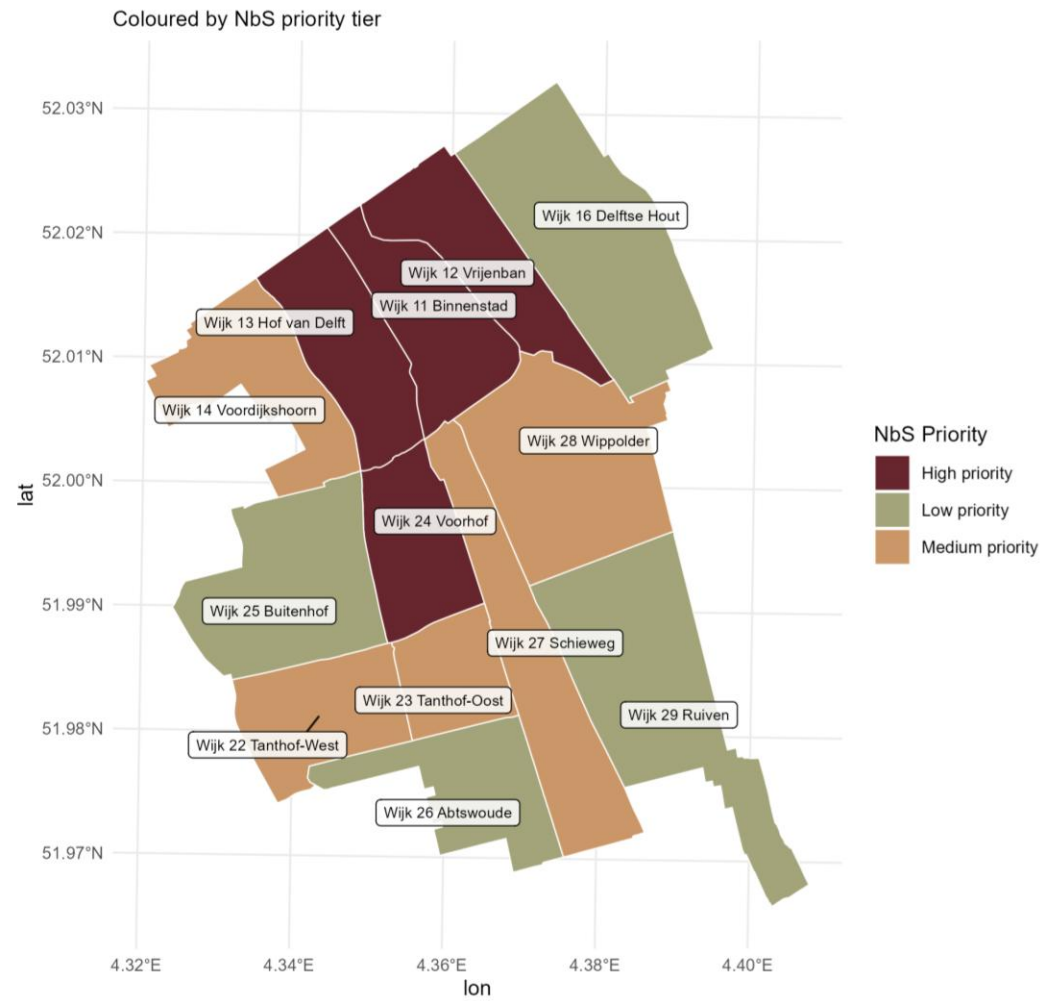
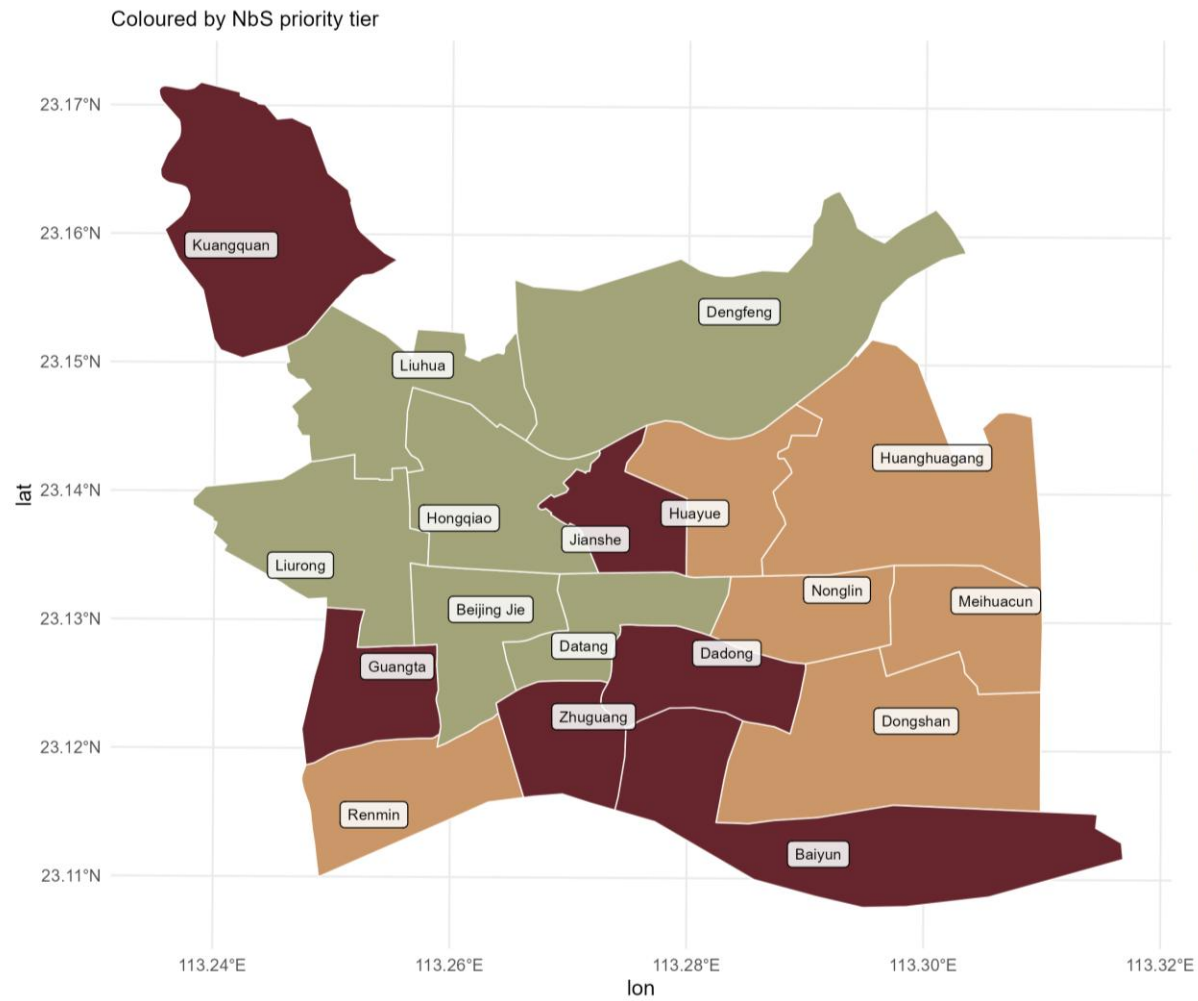
MCDA

NBS INTERVENTION POLICY



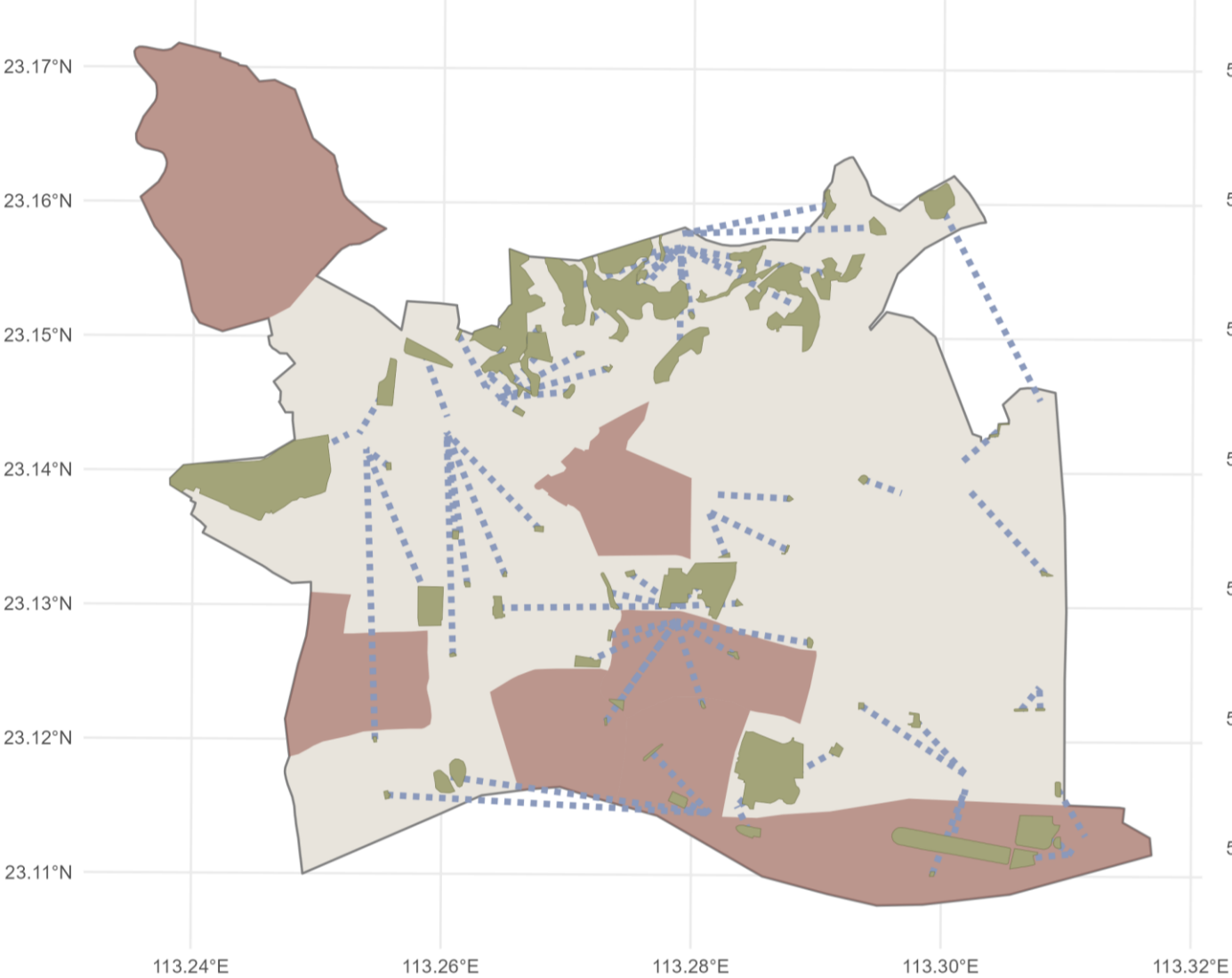
MCDA

NBS INTERVENTION POLICY WITH REFERENCE

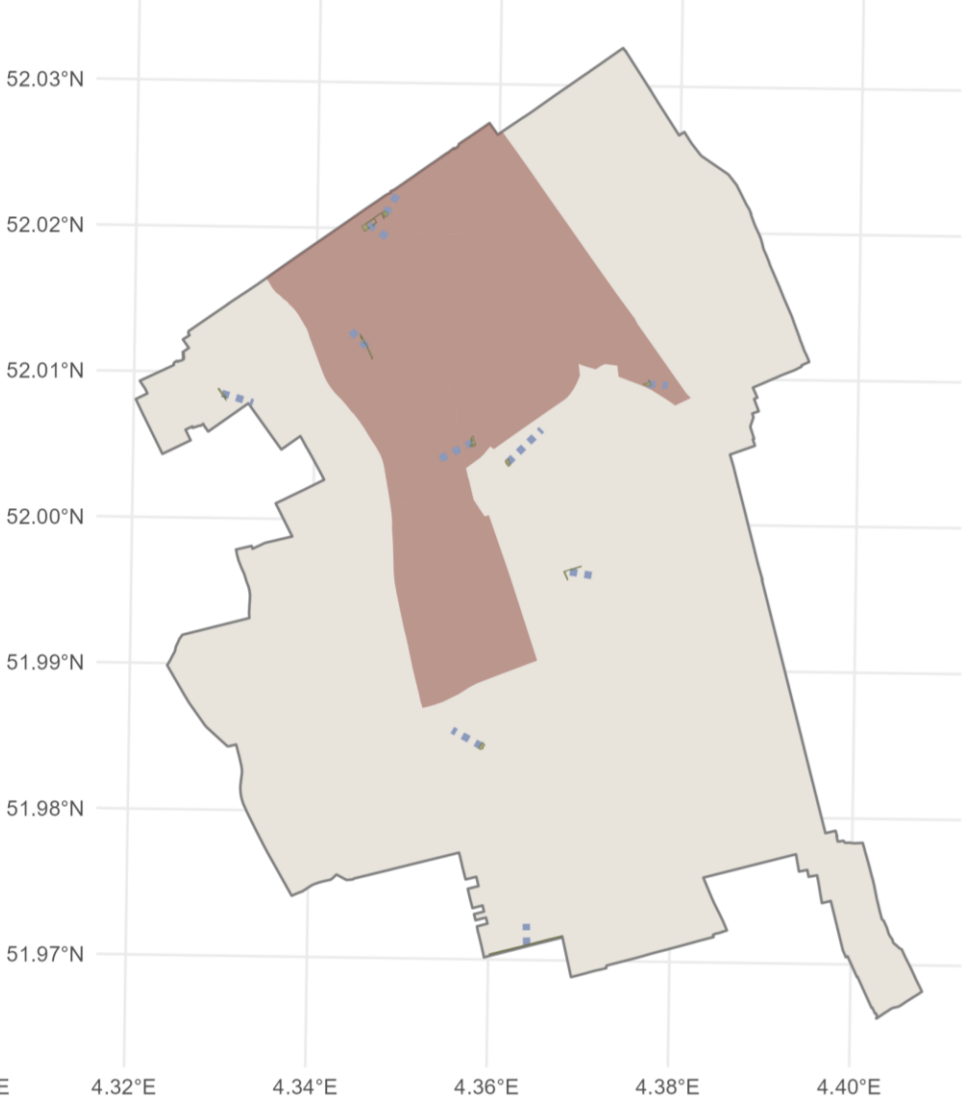


NBS CORRIDOR PRIORITISATION

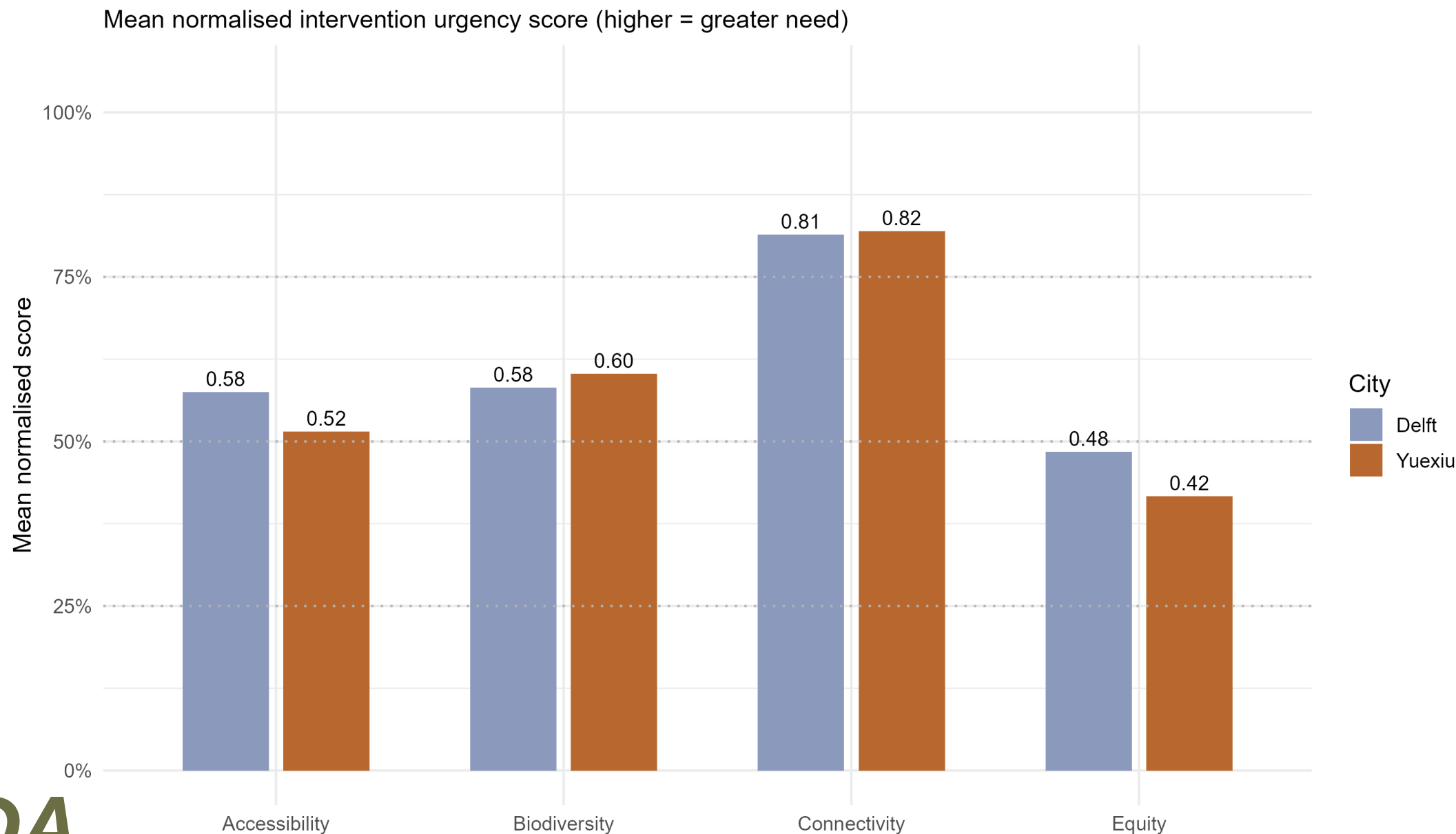
Red = high priority | Green = isolated patches | Dashed = proposed corridors



Red = high priority | Green = isolated patches | Dashed = proposed corridors



MCDA SUB-SCORE BAR CHART (city comparison)



Data source limitation

Scale and temporal comparability between the two cities

Flattening of variation due to chosen scale (neighborhood/sub-district)

Public dataset unavailability

LIMITATIONS

- **Replace degree-based connectivity with continuous dispersal kernel or betweenness centrality weighted by patch area**
- **Incorporate temporal dimension; green space change over time and projected population growth**
- **Apply symmetric wealth proxies across both cities for a robust equity comparison**
- **Extend NDVI analysis across seasons to capture vegetation stress patterns**
- **Test sensitivity of MCDA weights; current weighting is theoretically justified but alternative configurations should be explored**

FURTHER RESEARCH

**THANK
YOU!**

